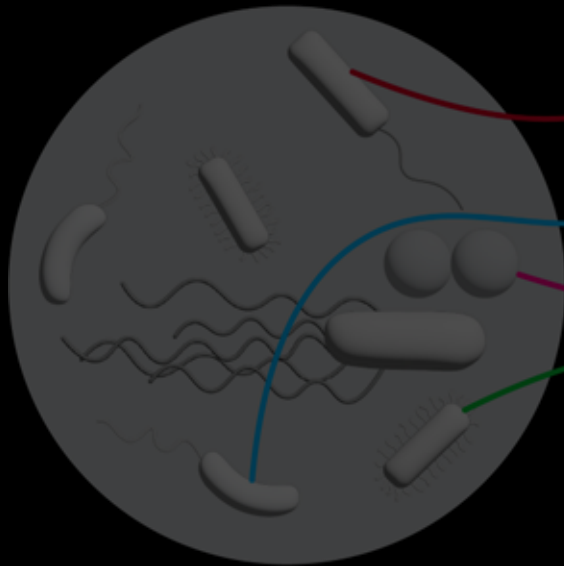


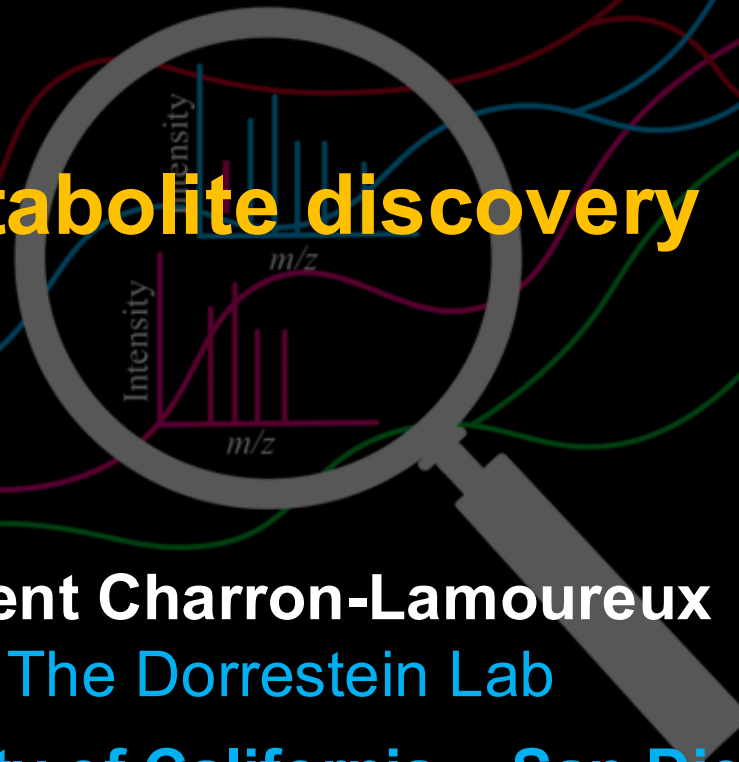


**Microbial
communities**

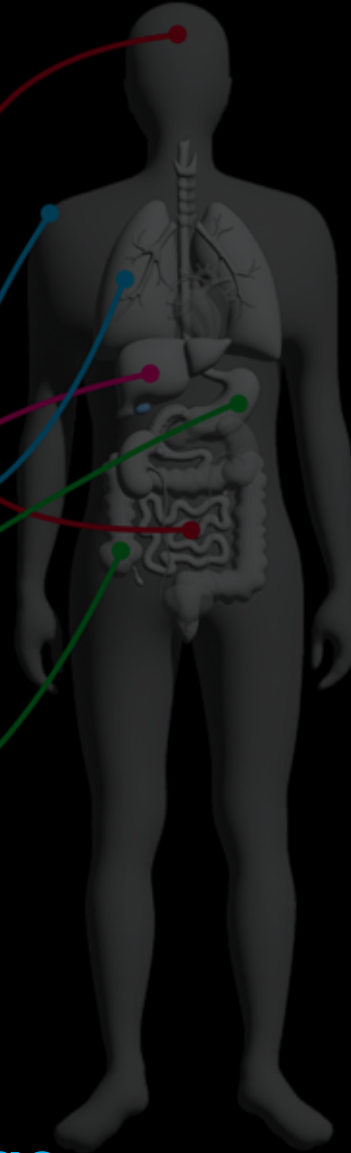


Mass spectrometry

Metabolite discovery



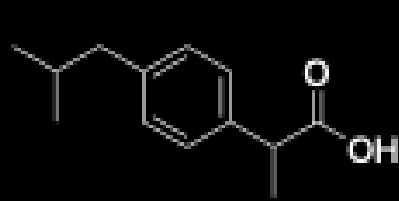
Host phenotypes



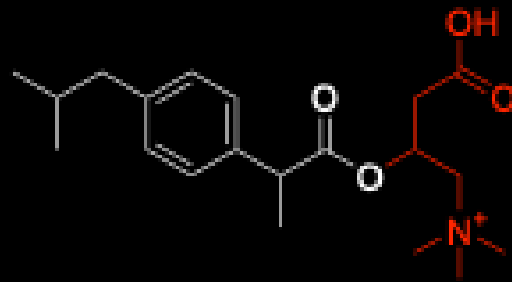
Vincent Charron-Lamoureux
The Dorrestein Lab

University of California – San Diego

Even the most studied drugs have unknown metabolites



Ibuprofen

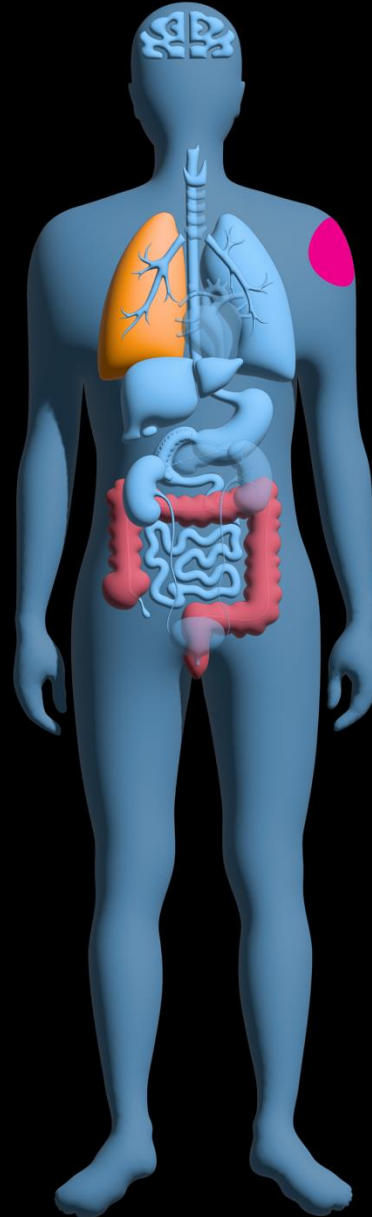
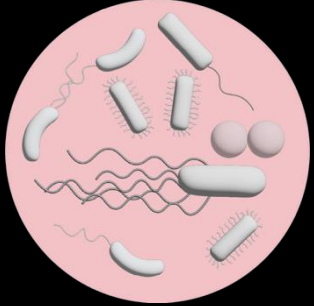


Ibuprofen-carnitine
(origin → still unknown)



Altered microbiota modulates human chemistry

Microbiota



**Biological context /
interventions**



Drugs (e.g.,
enalapril, carvediol)

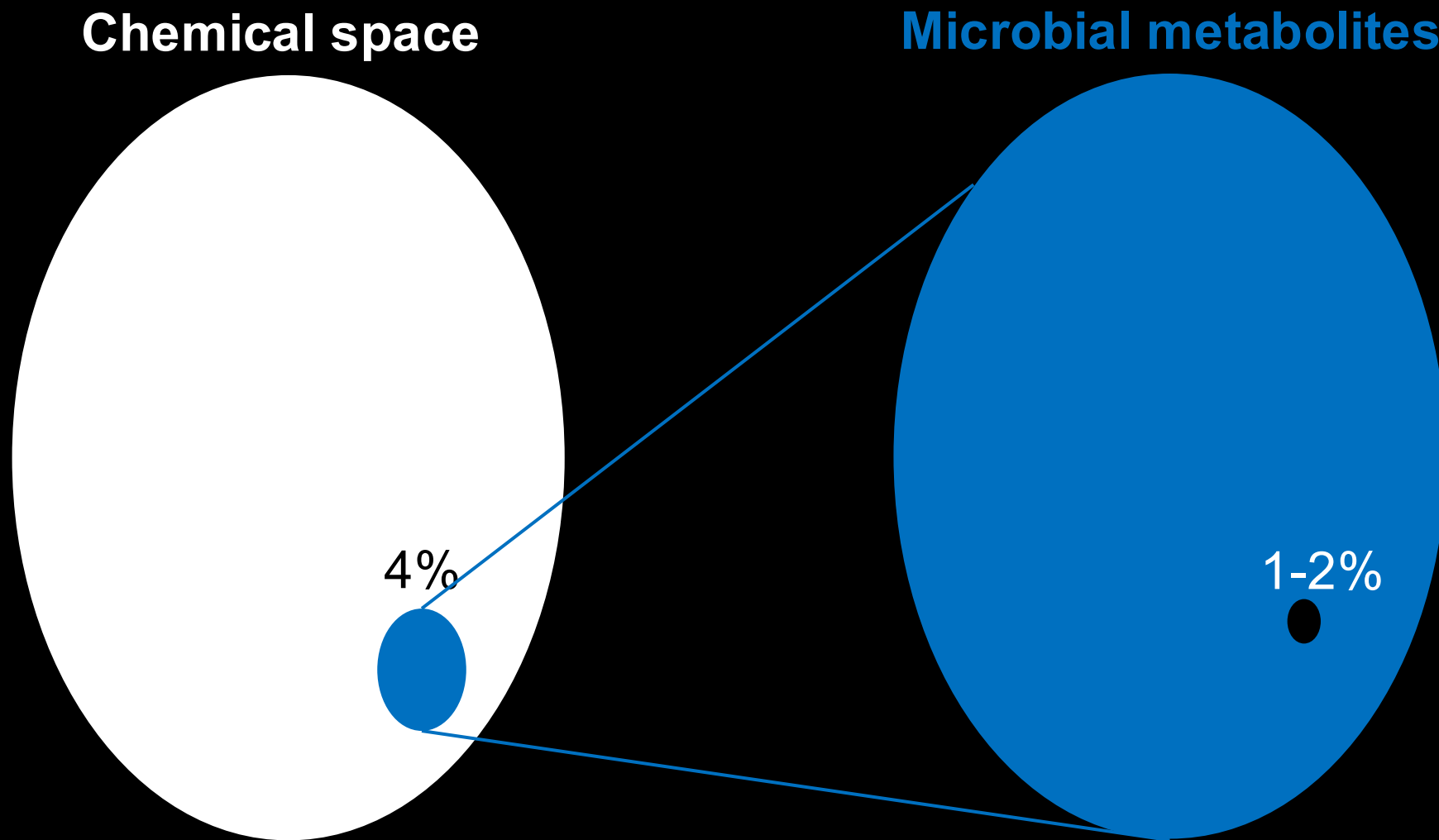


Health phenotypes
(e.g., IBD, Alzheimer's
disease)



Diets (e.g.,
HFD/LFD)

96% of the chemical space is still unknown



Untapped potential for small-molecule discovery

96% of the MS/MS spectra found in the public domain are still unannotated with a structure

What is an MS/MS spectrum?

The fingerprint of a molecule

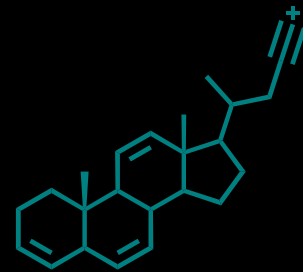
Four metabolomics repositories

GNPS/MassIVE

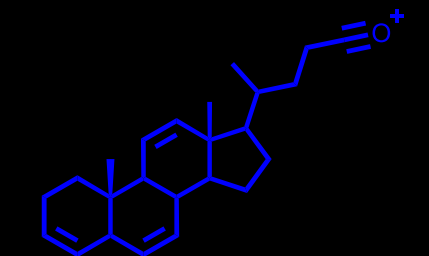
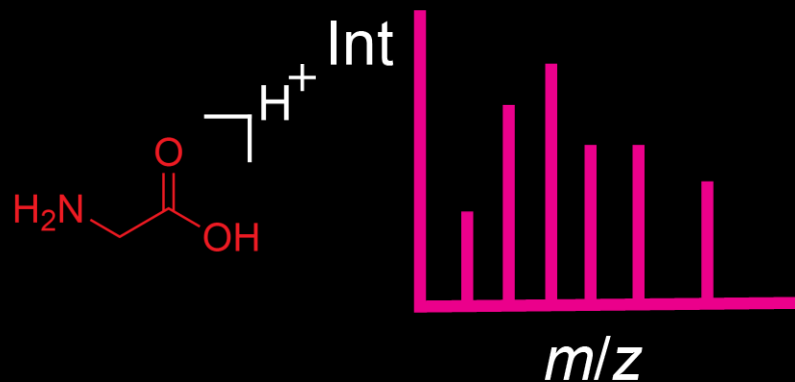
Metabolights

Metabolomics Workbench

NORMAN



Exact Mass: 319.24



Exact Mass: 337.25

Diagnostic ions

> 2 billion MS/MS spectra in the public domain

Impact of small molecules on human health

Bile acids and the gut microbiota: metabolic interactions and impacts on disease

Steph Secat Symb: [Shan G. Stine](#), [Jordan E. Bisanz](#), [C. Denise Okafor](#) & [Andrew D. Patterson](#) 

Qixing Ni et al. (2023) | [Cite this article](#)

Chenghao Zhao^{1,2} · Xi Luo^{1,2,16} · Meng Li^{1,2} · Kai Wang^{1,2,16}
Hua Lu⁸ · Yanli Pan¹ · Ming · Zhi ·

Published: 18 December 2021

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calorie restriction**
Yu Wang, [Shatin](#)

Qi Qu, Yan Chen, Yu
Yan-Hui Liu, Shengrong Xu, Cixiong
Chen, Shengye Xue, Baoding Zhang, Zhor
Guanghou Shui, ... Sheng-Cai Lin 

Crosstalk between in entero-hepatic in

Morgane M. Thibaut · Laure B. Bindels

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Intestinal crypt microbiota modulates intestinal stem cell turnover and tumorigenesis via indole acetic acid

[Shuning Zhang](#), [Lihua Peng](#), [Shyamal Goswami](#), [Yuchen Li](#), [Haiyue Dang](#), [Shuli Xing](#), [Panpan Feng](#),
[Giulia Nigro](#), [Yingying Liu](#), [Yingfei Ma](#), [Tianhao Liu](#), [Jiahua Yang](#), [Tinglei Jiang](#), [Yingnan Yang](#), [Nick
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RESOURCE · [Volume 188, Issue 15, P4154-4169.E](#)

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The microbiome diversity
fatty acid-derived *N*-acetyl

[Helena Mannocho-Russo](#)¹ · [Vincent Charron](#)
[Dionicio Siegel](#)¹ · [Rob Knight](#)^{6,13,23,24,25} · [Pia](#)

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Article | Published: 22 June 2022

The gut metabolite indole-3 propionate promotes nerve regeneration and repair

[abeth Serger](#), [Lucia Luengo-Gutierrez](#), [Jessica S. Chadwick](#), [Guiping Kong](#), [Luming Zhou](#), [Greg wford](#), [Matt C. Danzi](#), [Antonis Myridakis](#), [Alexander Brandis](#), [Adesola Temitope Bello](#), [Franziska ller](#), [Alexandros Sanchez-Vassopoulos](#), [Francesco De Virgiliis](#), [Phoebe Liddell](#), [Marc Emmanuel](#) nas, [Jessica Strid](#), [Sridhar Mani](#), [Dylan Dodd](#) & [Simone Di Giovanni](#) 

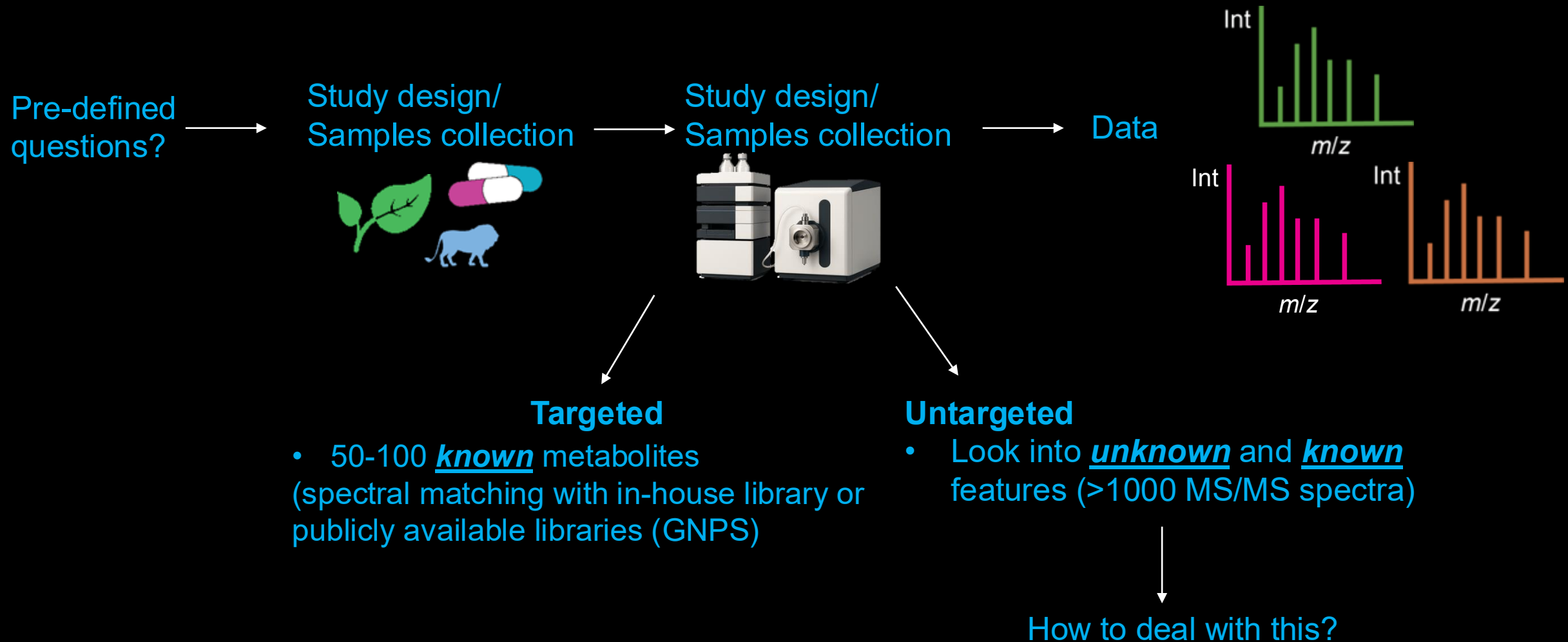
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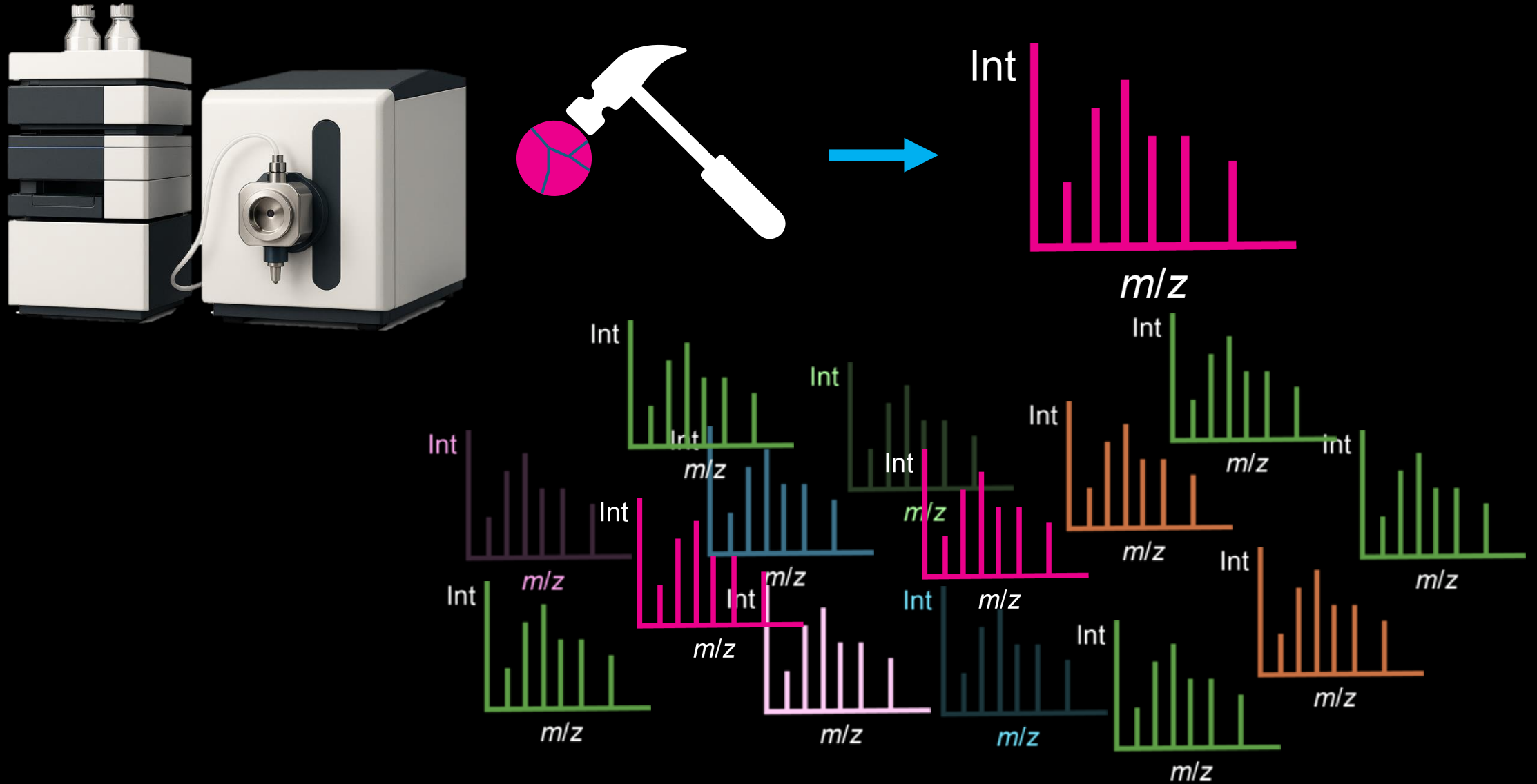
Antumor-specific T cell re-

Authors Info & Affiliations

Targeted vs untargeted metabolomics



How to find them – Untargeted metabolomics using LC-MS/MS



Untargeted LC-MS/MS generates a lot of data

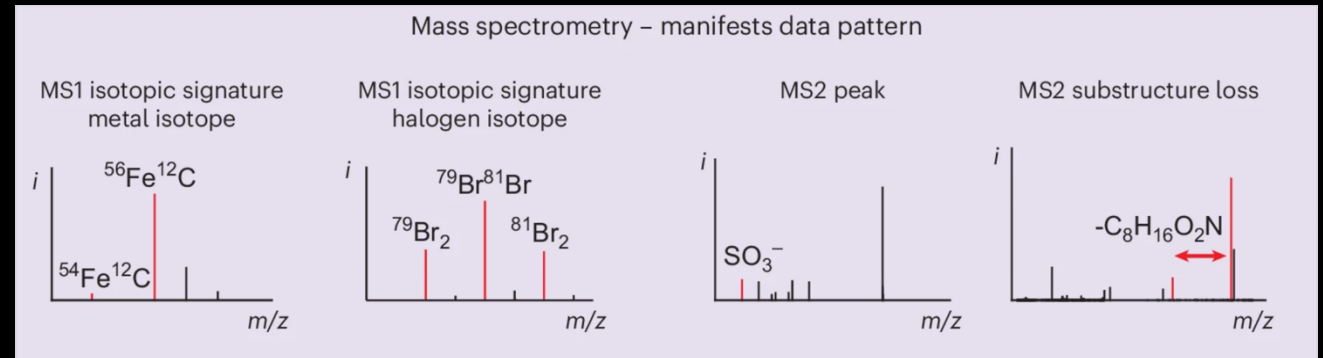
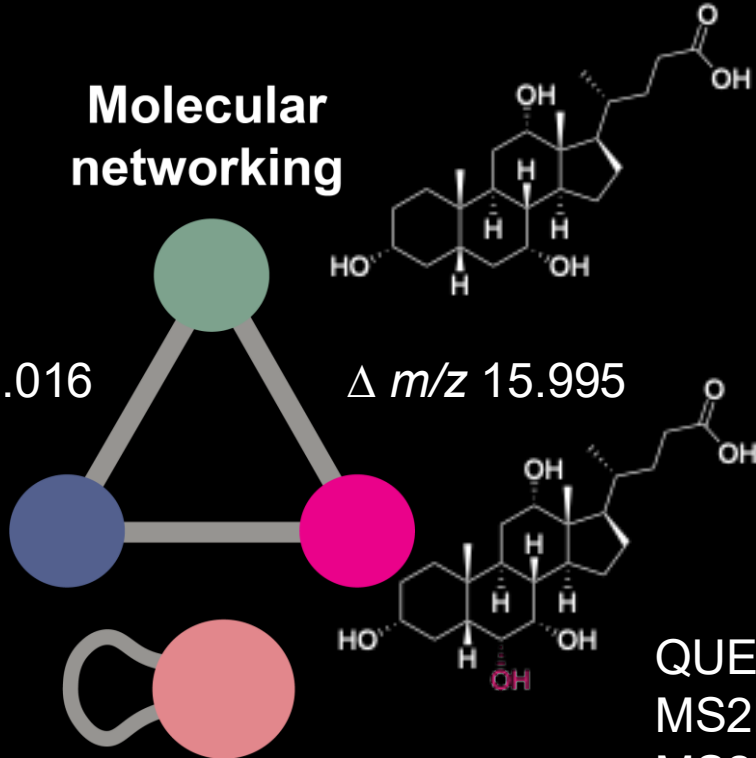
How to find them – with advanced computational tools



Molecular
networking

$\Delta m/z$ 14.016

$\Delta m/z$ 15.995



Mass Spec Query Language

```
QUERY scaninfo(MS2DATA) WHERE  
MS2PROD=337.25:TOLERANCEMZ=0.01:INTENSITYPERCENT=5 AND  
MS2PROD=319.24:TOLERANCEMZ=0.01:INTENSITYPERCENT=5
```

MassQL for discoveries – bile acids and *N*-acyl lipids



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The underappreciated diversity of bile acid modifications

[Ipsita Mohanty](#)^{1,18} · [Helena Mannocho-Russo](#)^{1,18} · [Joshua V. Schweer](#)^{1,17} · ... · [Mingxun Wang](#)⁷ · [Allegra T. Aron](#)¹⁰ · [Pieter C. Dorrestein](#)^{1,2,8,9,19} [✉](#)... [Show more](#)

→ > 21,000 bile acids

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The microbiome diversifies long- to short-chain fatty acid-derived *N*-acyl lipids

[Helena Mannocho-Russo](#)¹ · [Vincent Charron-Lamoureux](#)¹ · [Martijn van Faassen](#)^{1,2} · ... · [Dionicio Siegel](#)¹ · [Rob Knight](#)^{6,13,23,24,25} · [Pieter C. Dorrestein](#)^{1,13,26,27,28} [✉](#)... [Show more](#)

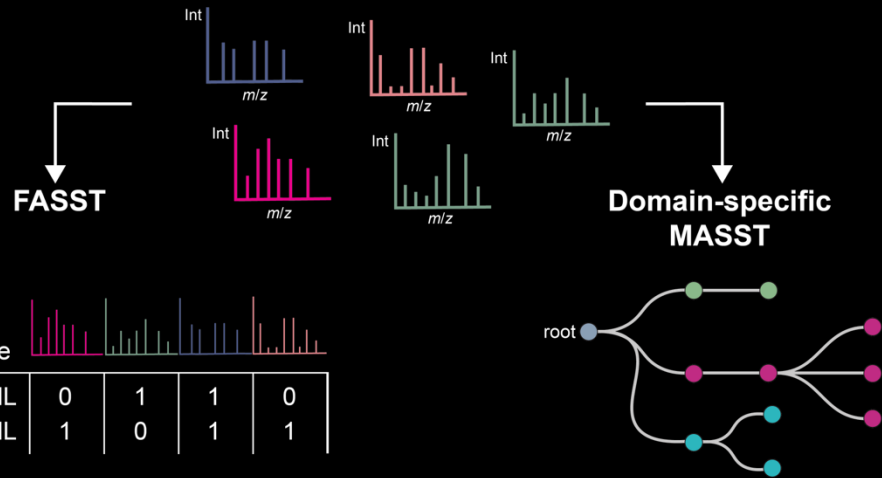
→ > 800 *N*-acyl lipids

How to find them – Contextualization workflow

Reverse metabolomics

Part 1

MS/MS spectra collection



Part 2

MassIVE:filename	MS/MS spectra			
PublicationA.mzML	0	1	1	0
PublicationB.mzML	1	0	1	1

Part 3

Repository-scale Phenotypic association



Interventions
Drugs, probiotics, diets

Organ distribution

Life stage

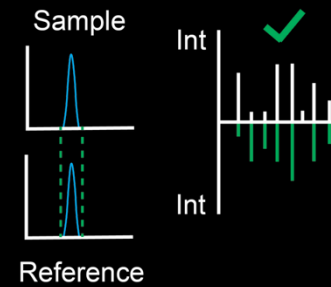
Diseases
COVID-19, Cystic fibrosis, Cirrhosis



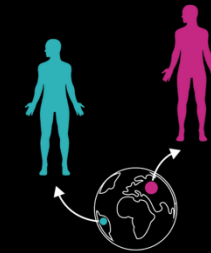
Part 4

Validation steps

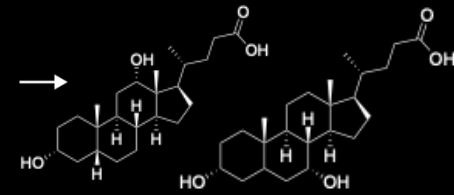
Retention time and MS/MS spectral matching



Different cohorts



Different isomers



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This Week's Featured Protocol:

A guide to reverse metabolomics

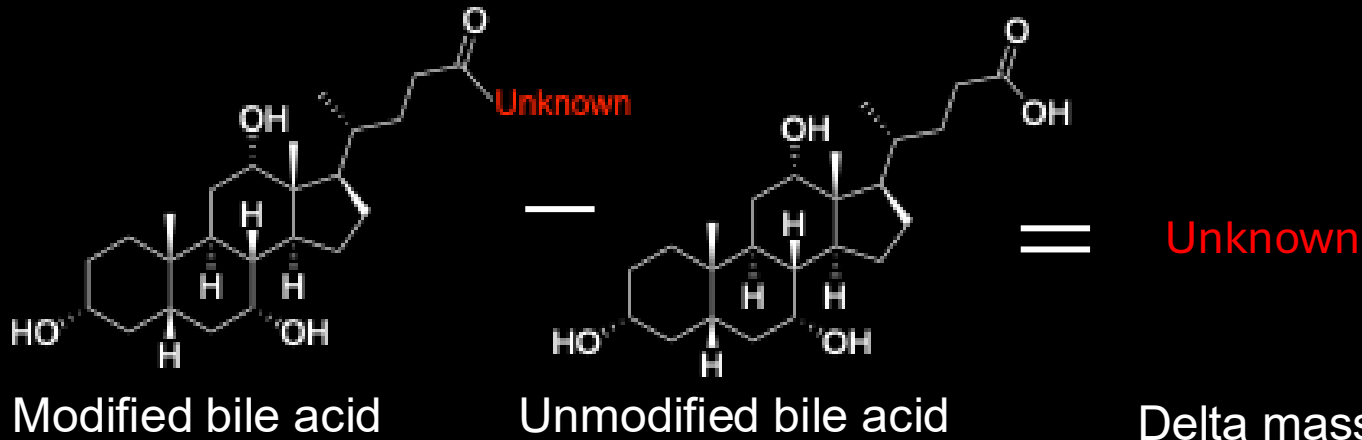
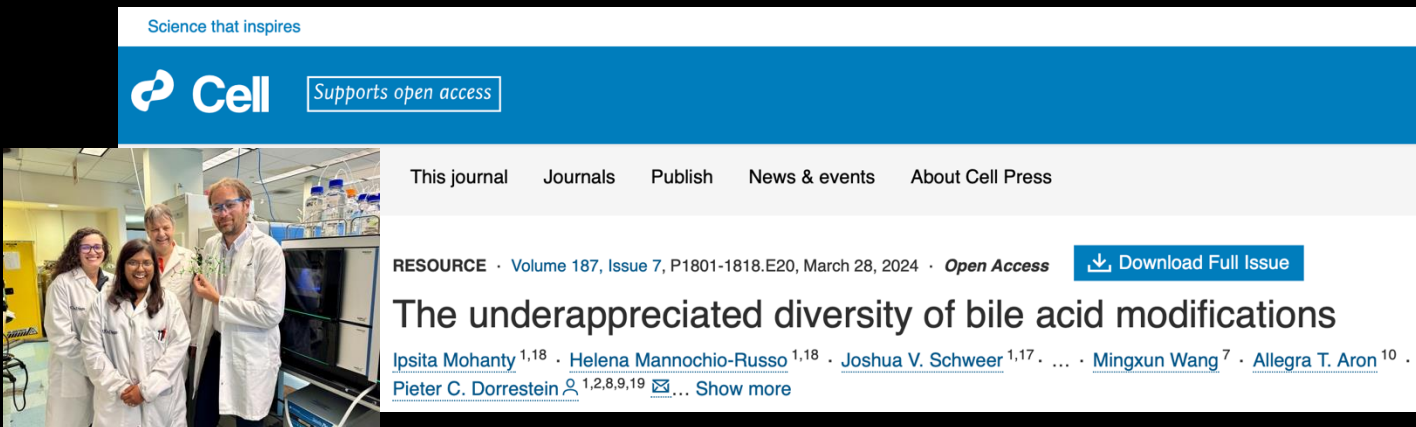
Featured

Questions?

A candidate bile acid library from public data

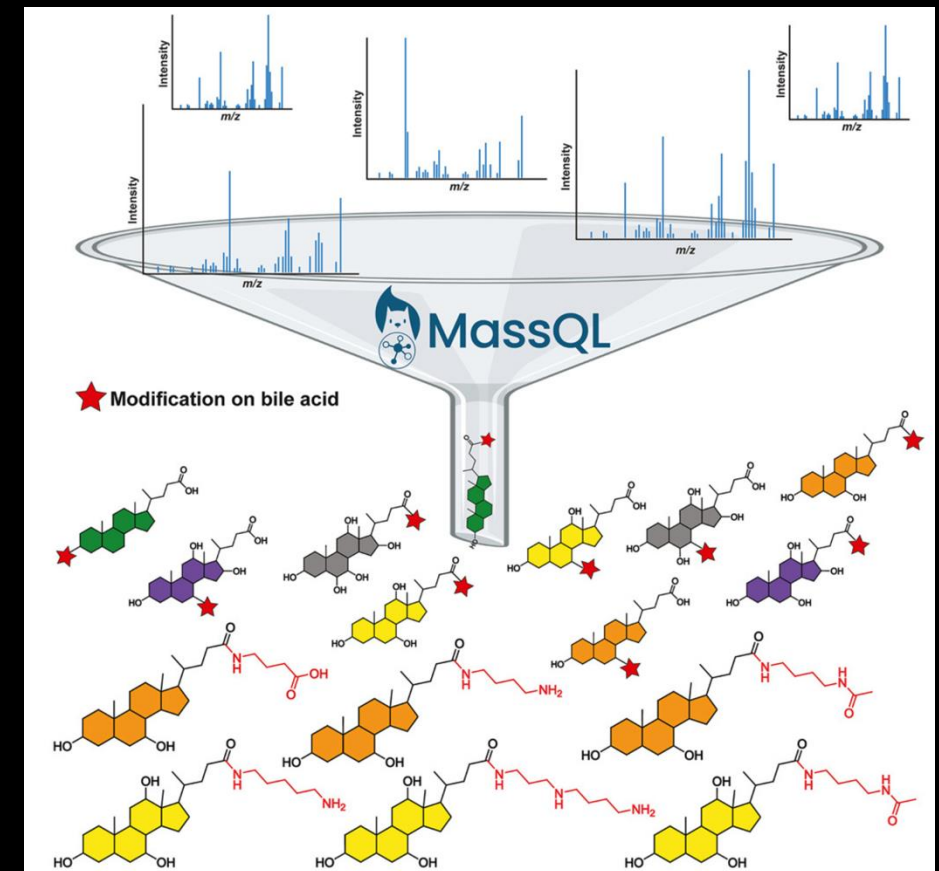
Bile acids

- Signalling molecules
- Involved in aging
- Participate in immune regulation
- Diseases (e.g., MASH, IBD, Cancer)

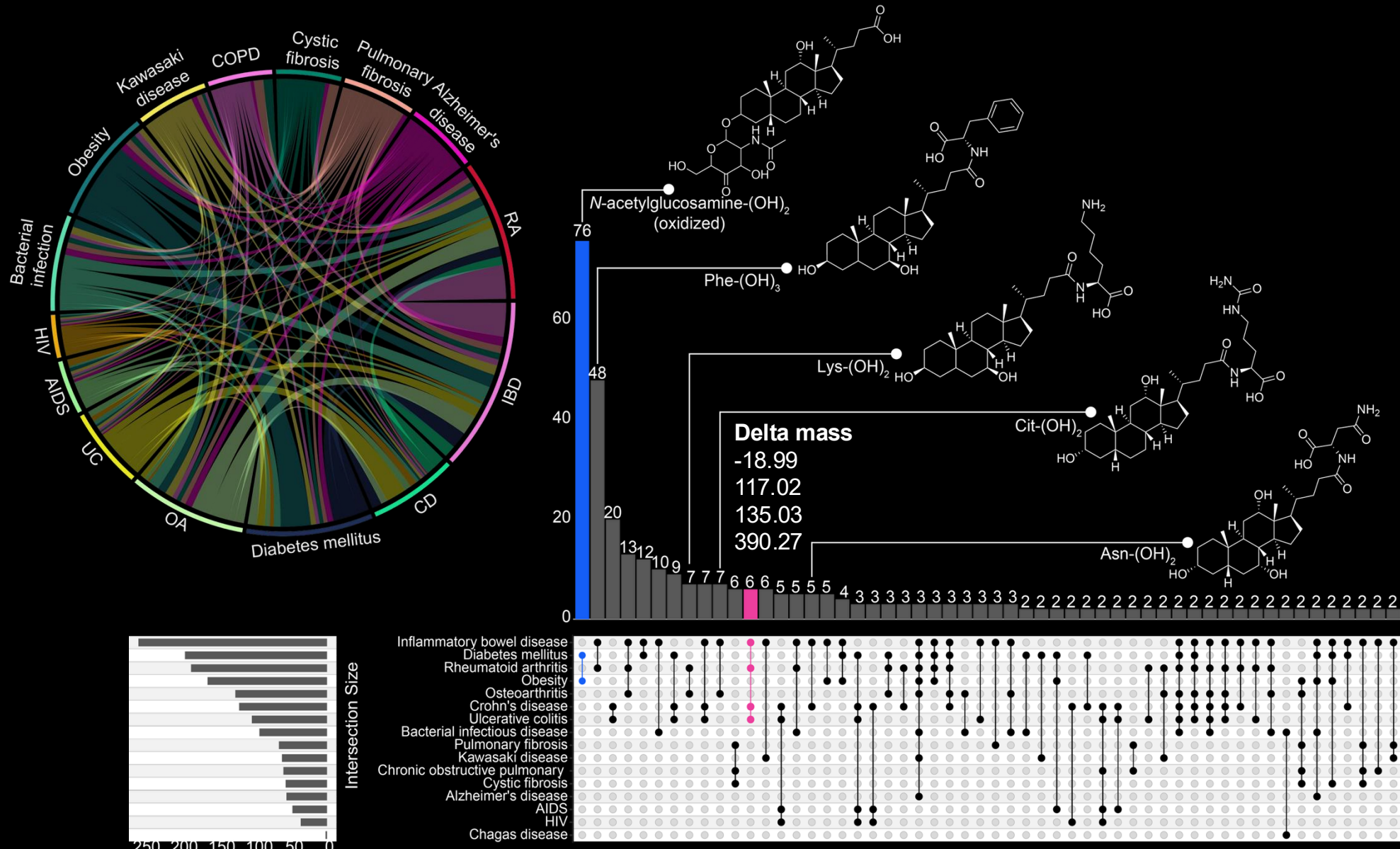


Public data mining

- 21,000 candidate bile acids (library)



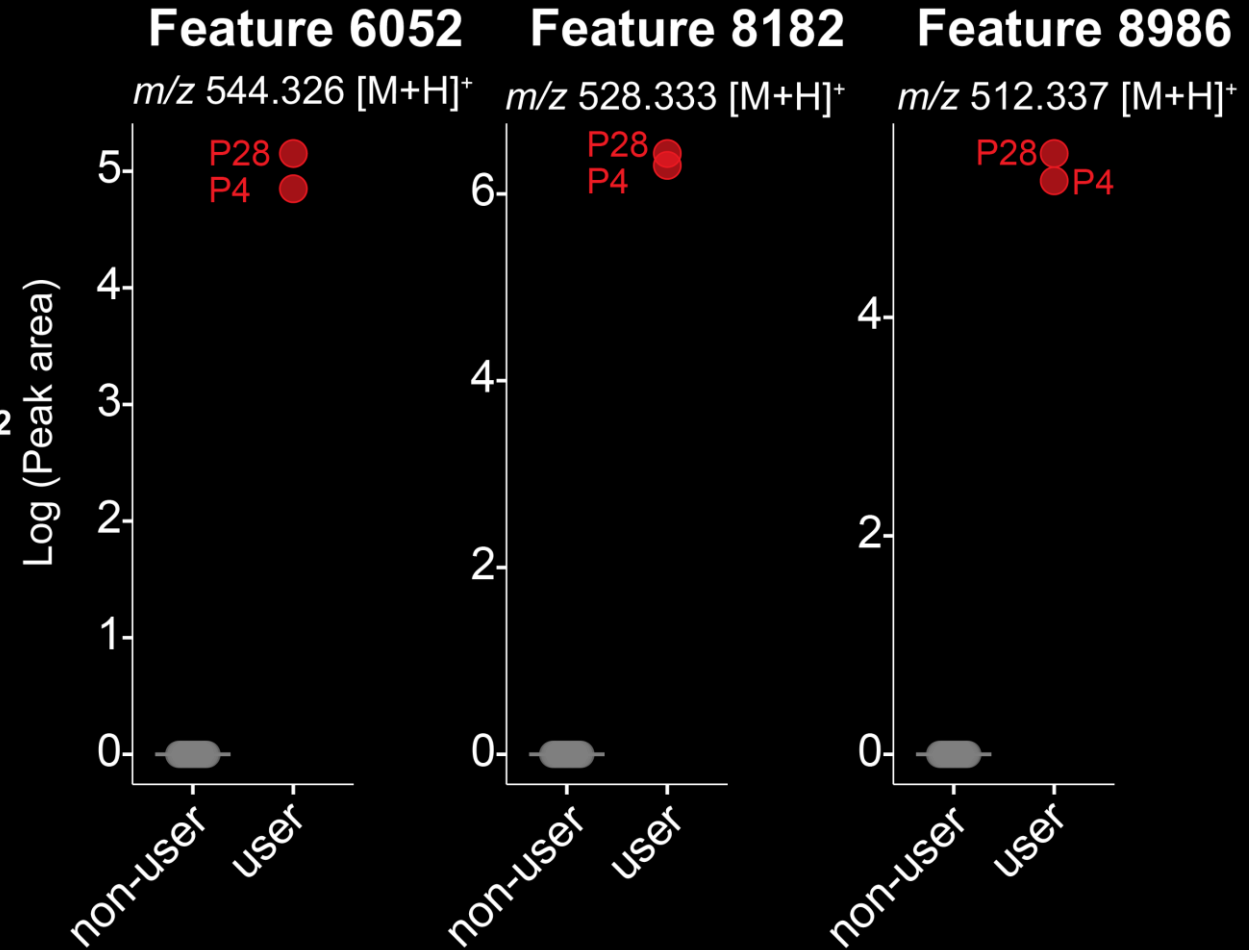
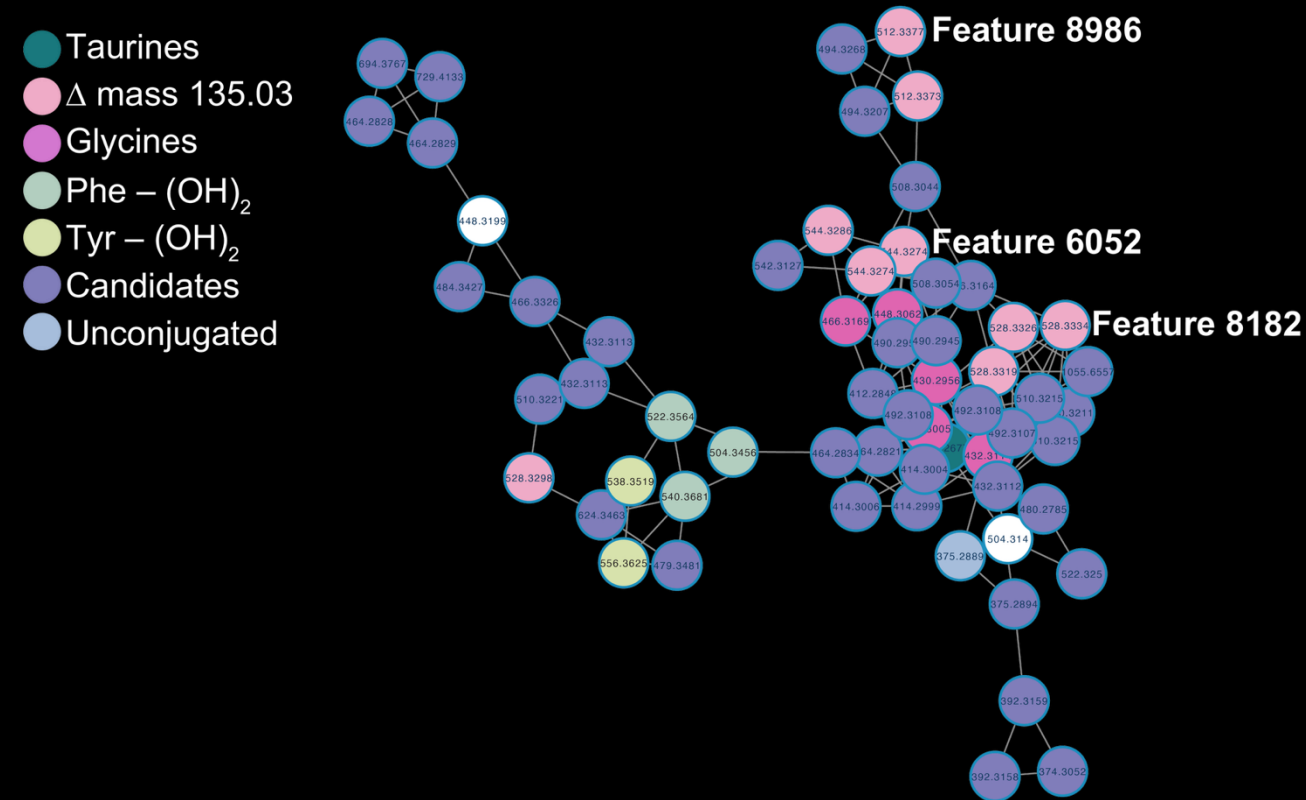
Finding disease association of unknown bile acids with reverse metabolomics



Unknown bile acids were found in two patients with rheumatoid arthritis

Reanalysis of public data

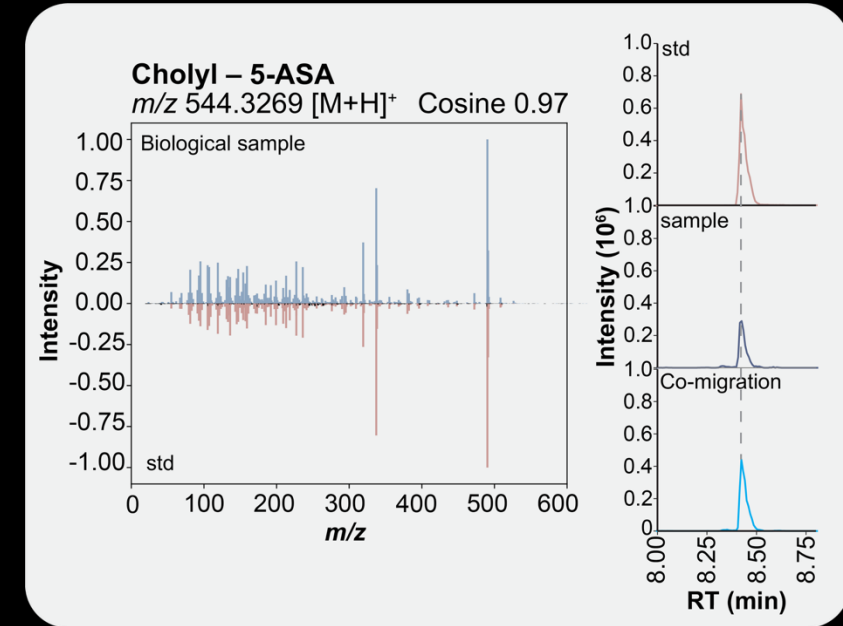
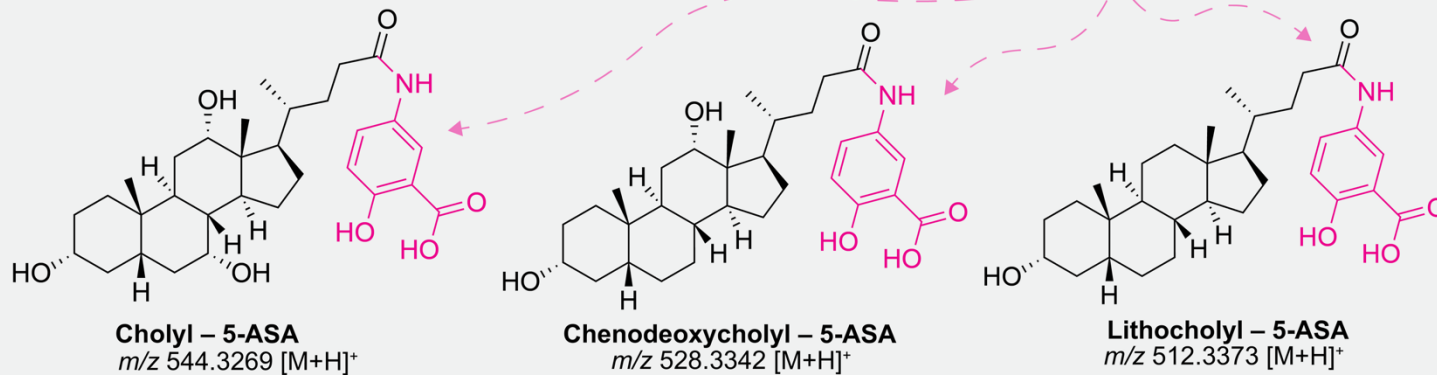
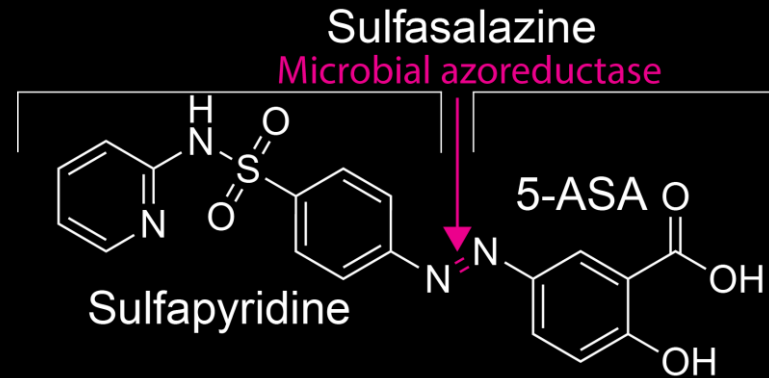
Molecular network
Rheumatoid arthritis cohort



Sulfasalazine

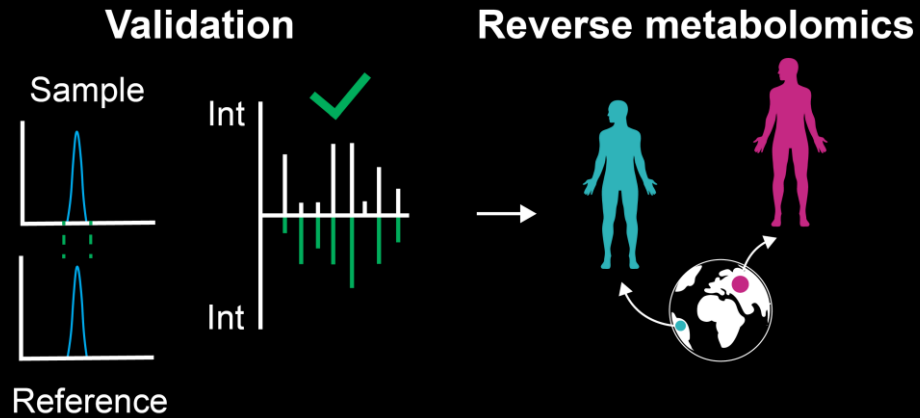
Reverse metabolomics identifies 5-ASA – bile acid conjugates

Pilot study Rheumatoid arthritis cohort



Low number of patients limits further analysis (e.g., responders vs non-responders)

Cholyl-5-ASA is found across multiple IBD cohorts



Cross-study molecular network analysis reveals dozens of new 5-ASA derivatives

naturemedicine

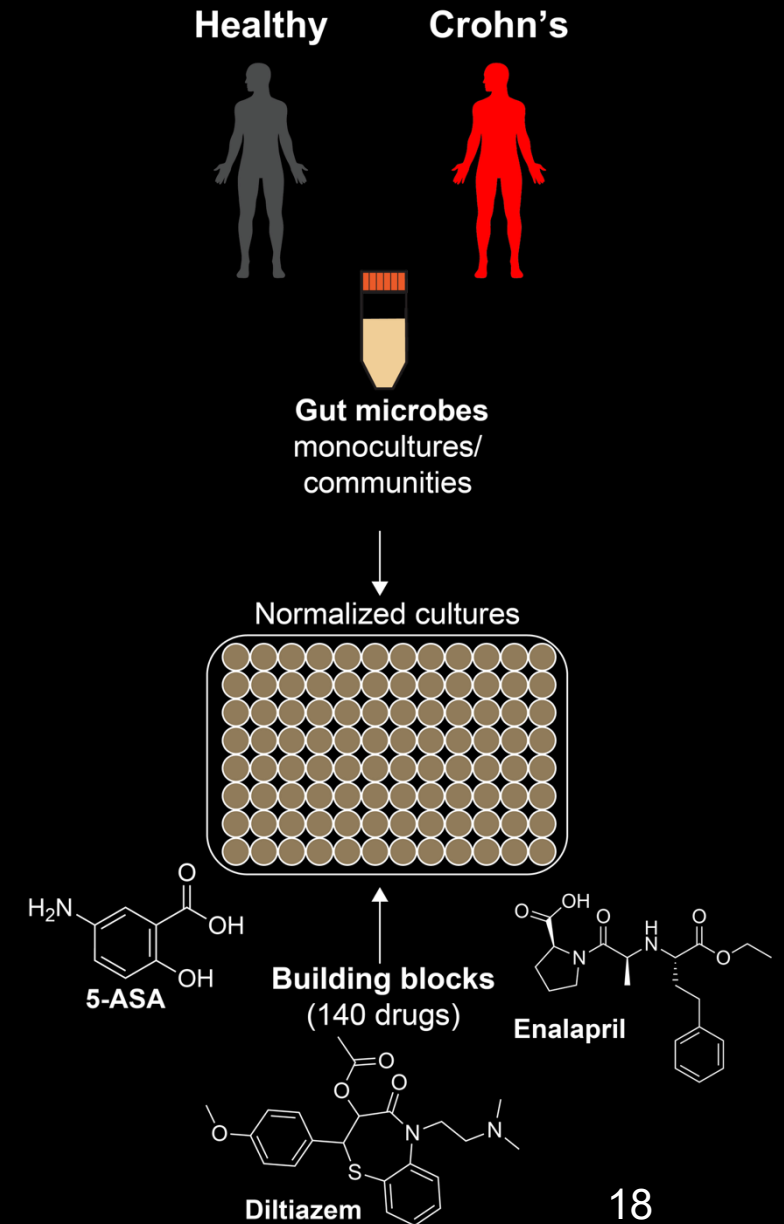
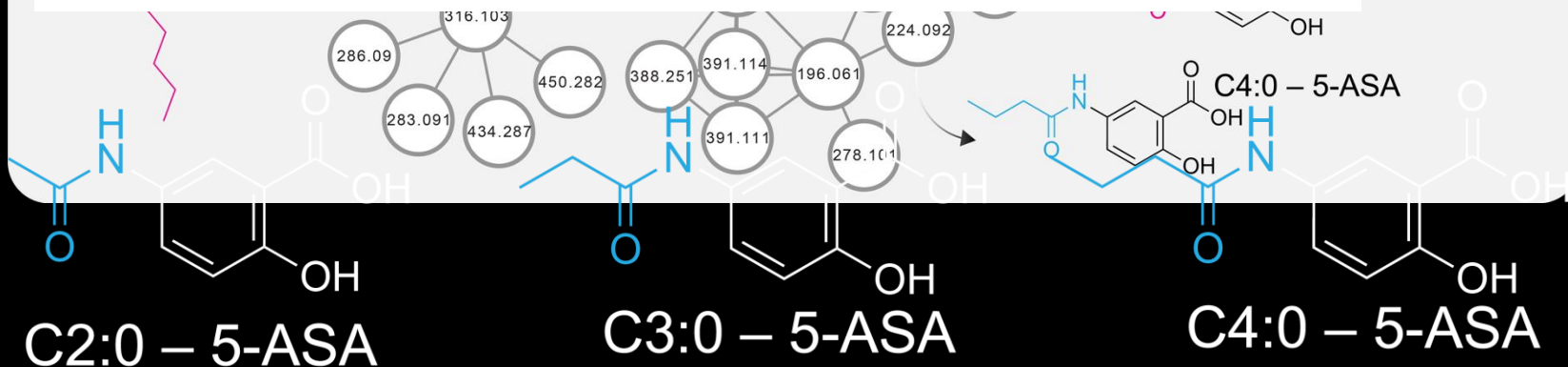
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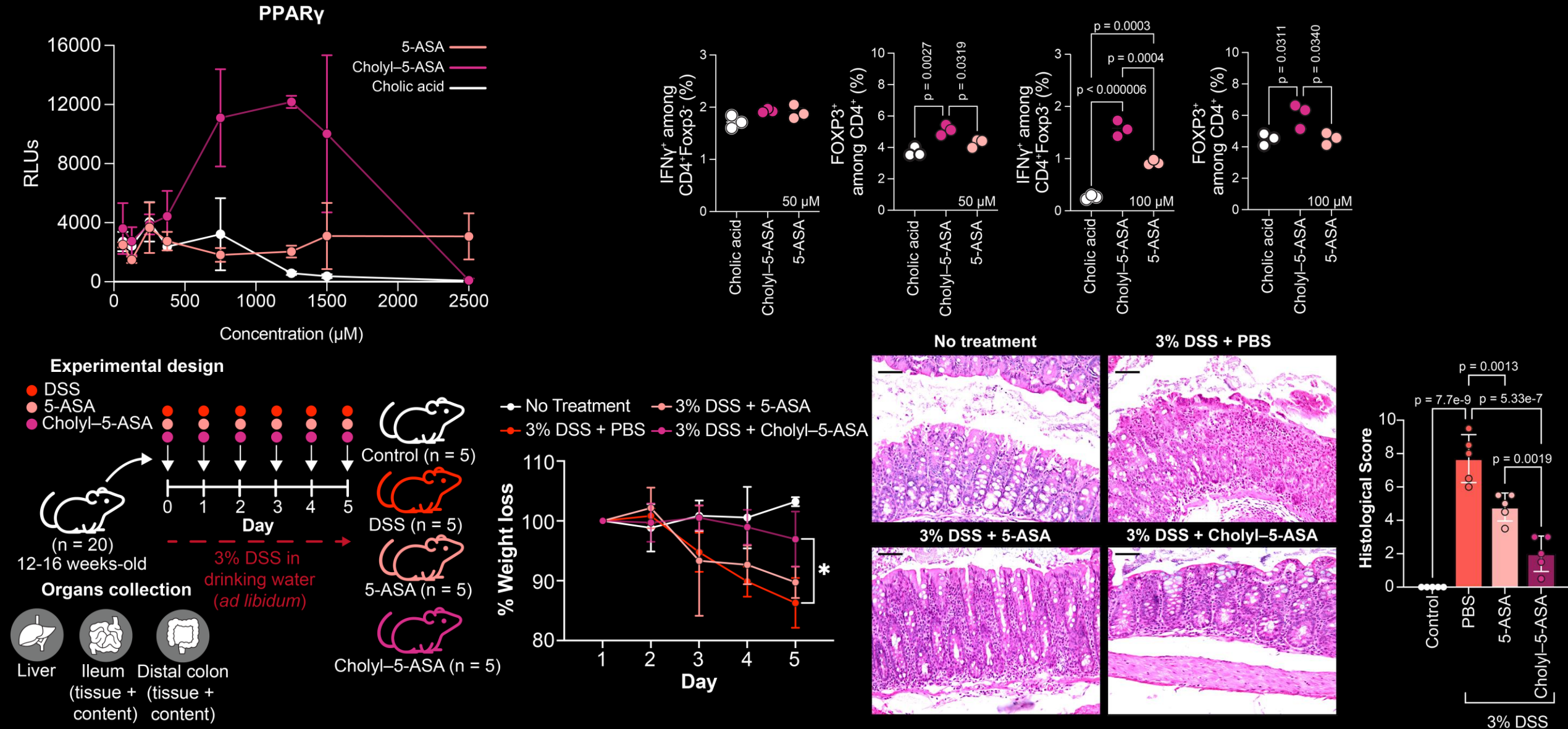
Gut microbial metabolism of 5-ASA diminishes its clinical efficacy in inflammatory bowel disease

[Raaj S. Mehta](#), [Jared R. Mayers](#), [Yancong Zhang](#), [Amrisha Bhosle](#), [Nathaniel R. Glasser](#), [Long H. Nguyen](#), [Wenjie Ma](#), [Sena Bae](#), [Tobyn Branck](#), [Kijun Song](#), [Luke Sebastian](#), [Julian Avila Pacheco](#), [Hyuk-Soo Seo](#), [Clary Clish](#), [Sirano Dhe-Paganon](#), [Ashwin N. Ananthakrishnan](#), [Eric A. Franzosa](#), [Emily P. Balskus](#), [Andrew T. Chan](#) & [Curtis Huttenhower](#) ✉

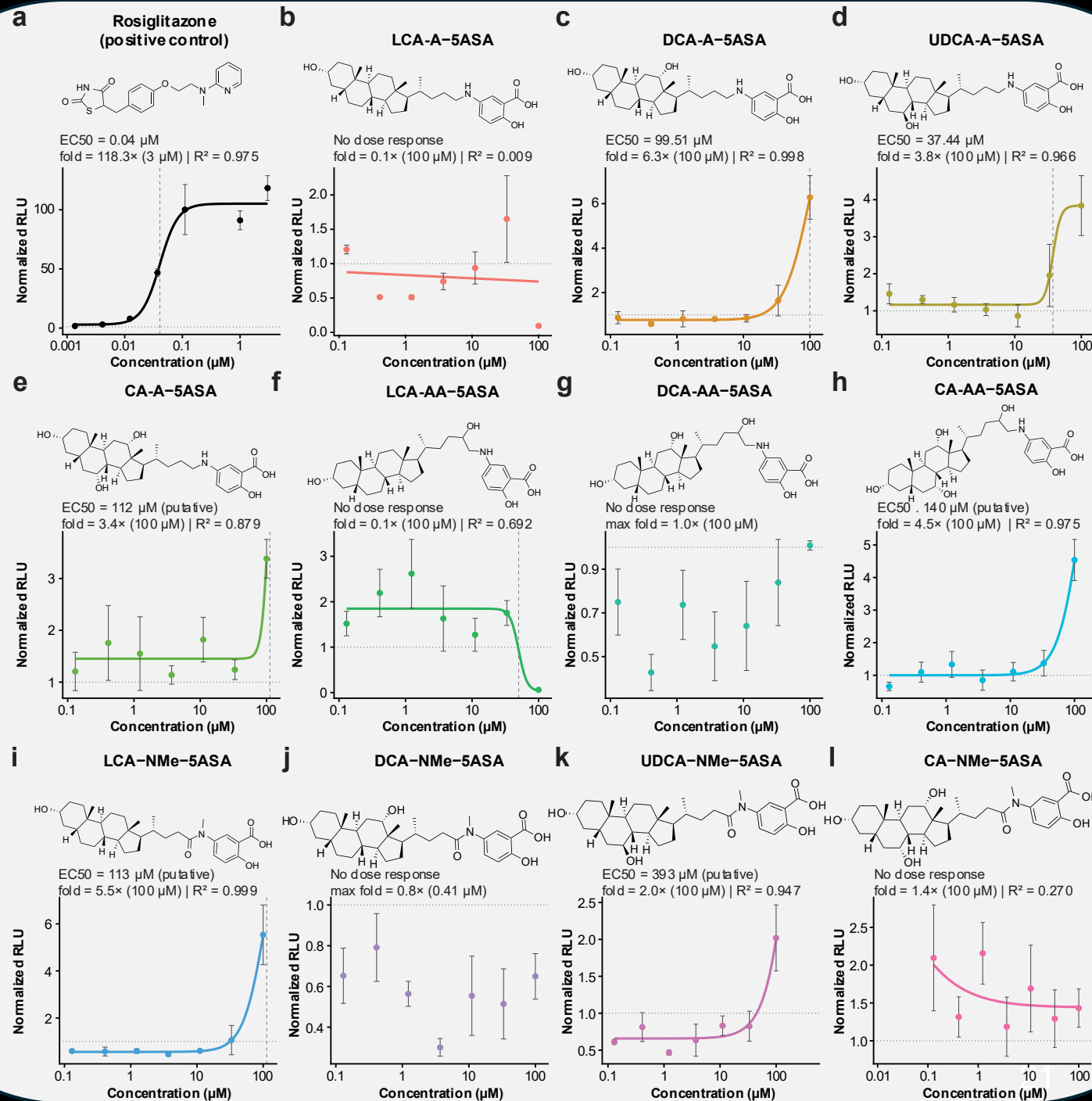


Charron-Lamoureux, V., [...], & Dorrestein, PC. Pan-repository analysis reveals a drug-activating function of microbial bile acid conjugation. *bioRxiv* (2026) under review

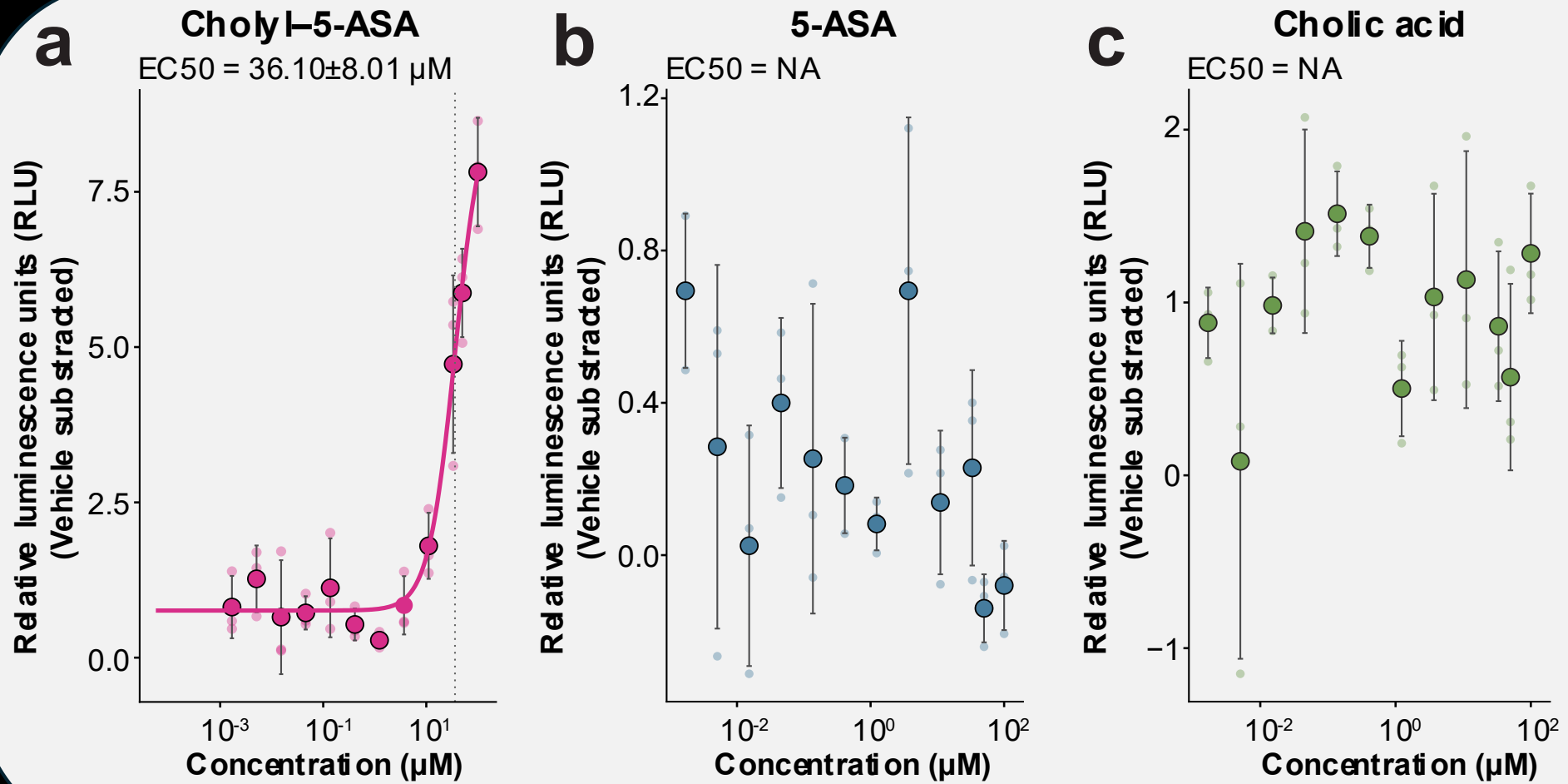
Drug-activating function of microbial bile acid conjugation



Prodrug hypothesis? – non-hydrolysable conjugates



CholyI-5-ASA stimulates AhR receptor



Why is the microbiome difficult to study?

Multiple interacting variables



5-ASA

Drugs (e.g., enalapril, fluoxetine)



Health phenotypes (e.g., IBD,
Alzheimer's disease)



Dietary (e.g., HFD/LFD)

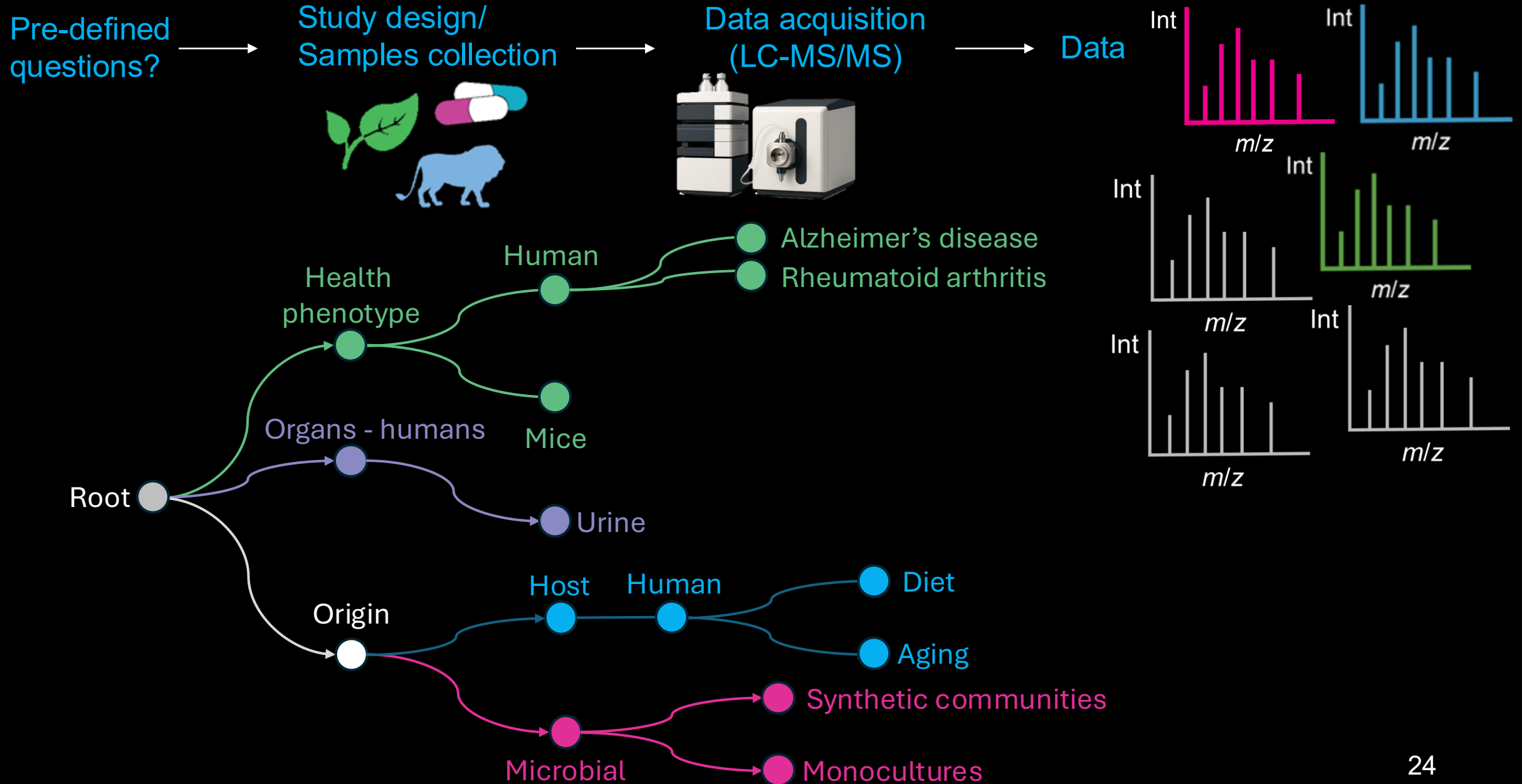


Fragmented, study-by-study

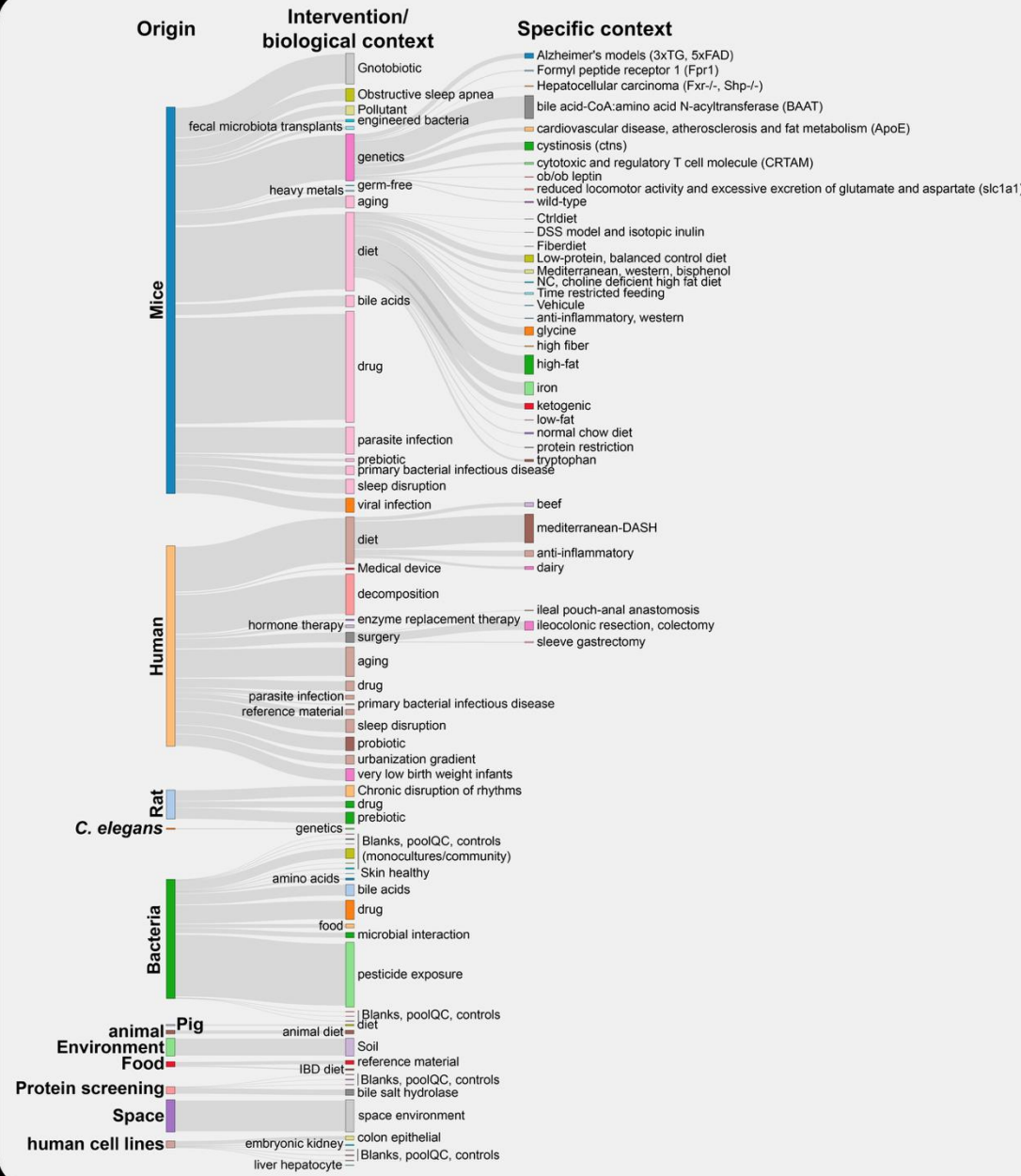
- Limited to individual studies
- Difficult to assess broader biological relevance

Can we create a resource to search all these studies simultaneously?

How we build a microbiome-centric network graph



How we build a microbiome-centric network graph



MicrobiomeMASST

- 467 studies
- 144,553 LC-MS runs

Biological interventions

- Human diets – 1,812 LC-MS files
- Human aging – 1,201 LC-MS files
- Mice drugs – 3,822 LC-MS files
- Mice diets – 3,563 LC-MS files

What can we learn from a newly-discovered polyamine bile acid

Bile acids and the gut microbiota: metabolic interactions and impacts on disease

Stephanie

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The underappreciated diversity of bile acid modifications

Ipsita Mohanty^{1,18} · Helena Mannocho-Russo^{1,18} · Joshua V. Schweer^{1,17} · ... · Mingxun Wang⁷ · Allegra T. Aron¹⁰ · Pieter C. Dorrestein^{1,2,8,9,19} Show more

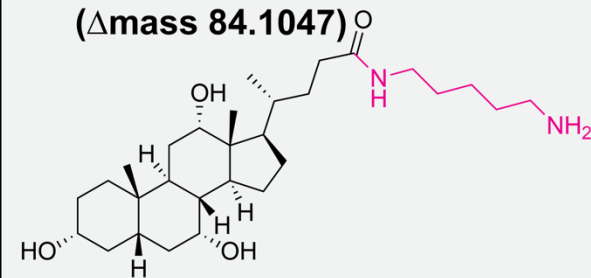
Affiliations & Notes Article Info

Gut symbionts alleviate MASH through a secondary bile acid biosynthetic pathway

Qixing Nie^{1,2,3,16} · Xi Luo^{1,2,16} · Kai Wang^{2,4,16} · Yong Ding^{1,2,16} · Shumi Jia^{5,16} · Qixiang Zhao^{1,2} · Meng Li^{1,2} · Jinxin Zhang^{1,2} · Yingying Zhuo^{1,2} · Jun Lin^{1,2} · Chenghao Guo^{1,2} · Zhiwei Zhang^{1,2} · Huiying Liu^{1,2} · Guangyi Zeng^{1,2} · Jie You⁶ · Lulu Sun^{1,7} · Yang, Mengqi Li, Xiao Tian, Xiaoyan Wei, Hua Lu⁸ · Ming Ma⁵ · Yanxing Jia⁵ · Ming-Hua Zheng^{9,10,11} · Yanli Pang^{1,12,13,14} · Jie Qiao^{1,12,13,14} · Changtao Jiang^{1,2,4,15,17} Show less

Chen, Shengye Xue, Baoding Zhang, Zhong-Zheng Zheng, Hai-Long Piao, Changtao Jiang, Hao Guo, Guanghou Shui, ... Sheng-Cai Lin Show authors

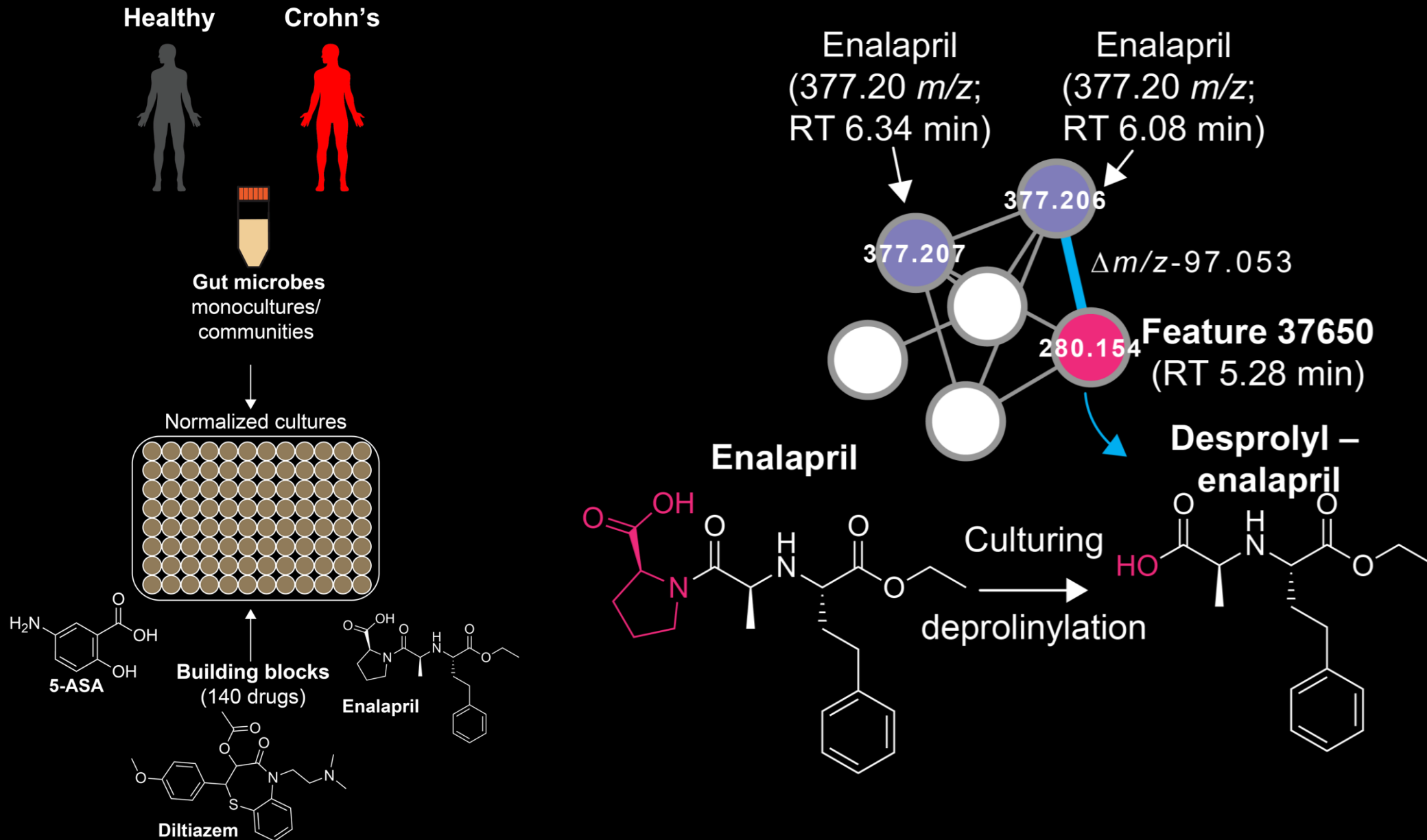
Trihydroxylated – cadaverine (Δ mass 84.1047)



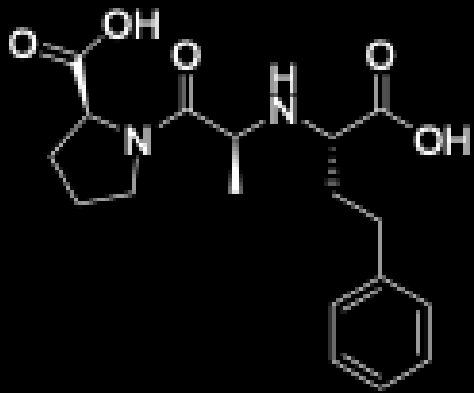
animal

9642|Ursus thibetanus
9689|Panthera leo
9694|Panthera tigris

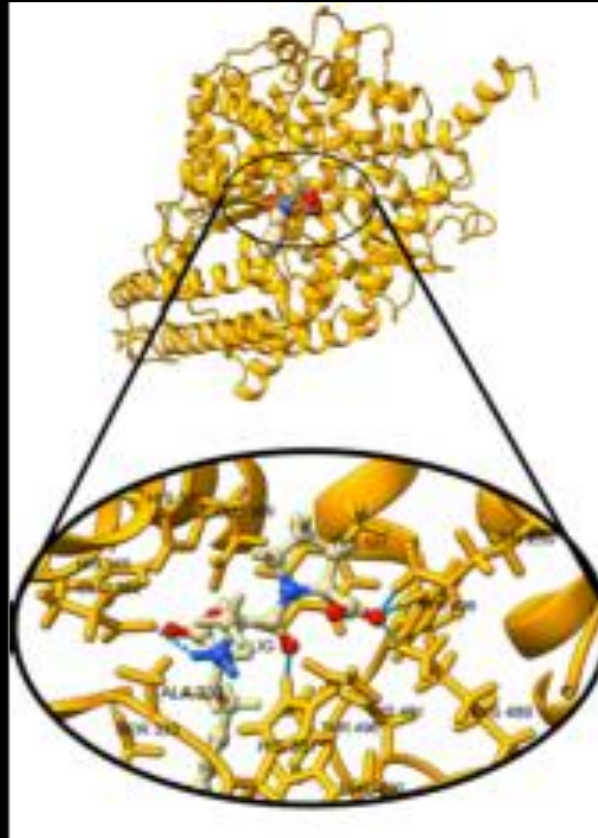
Screening of 164 drugs integrated within microbiomeMASST



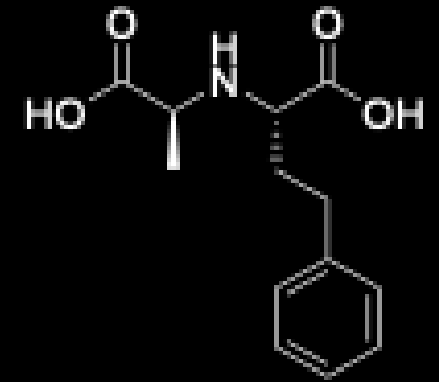
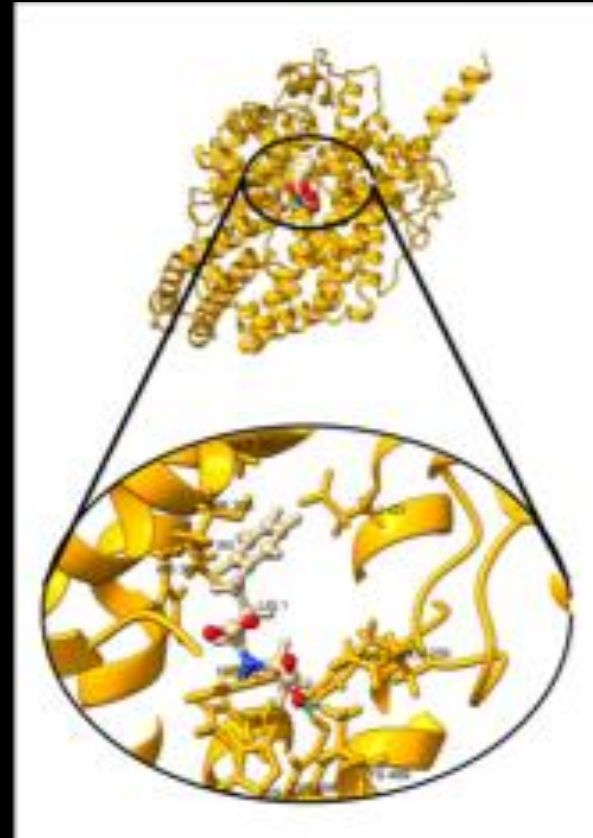
In silico binding assay using Boltz-2



Enalaprilat
(active)



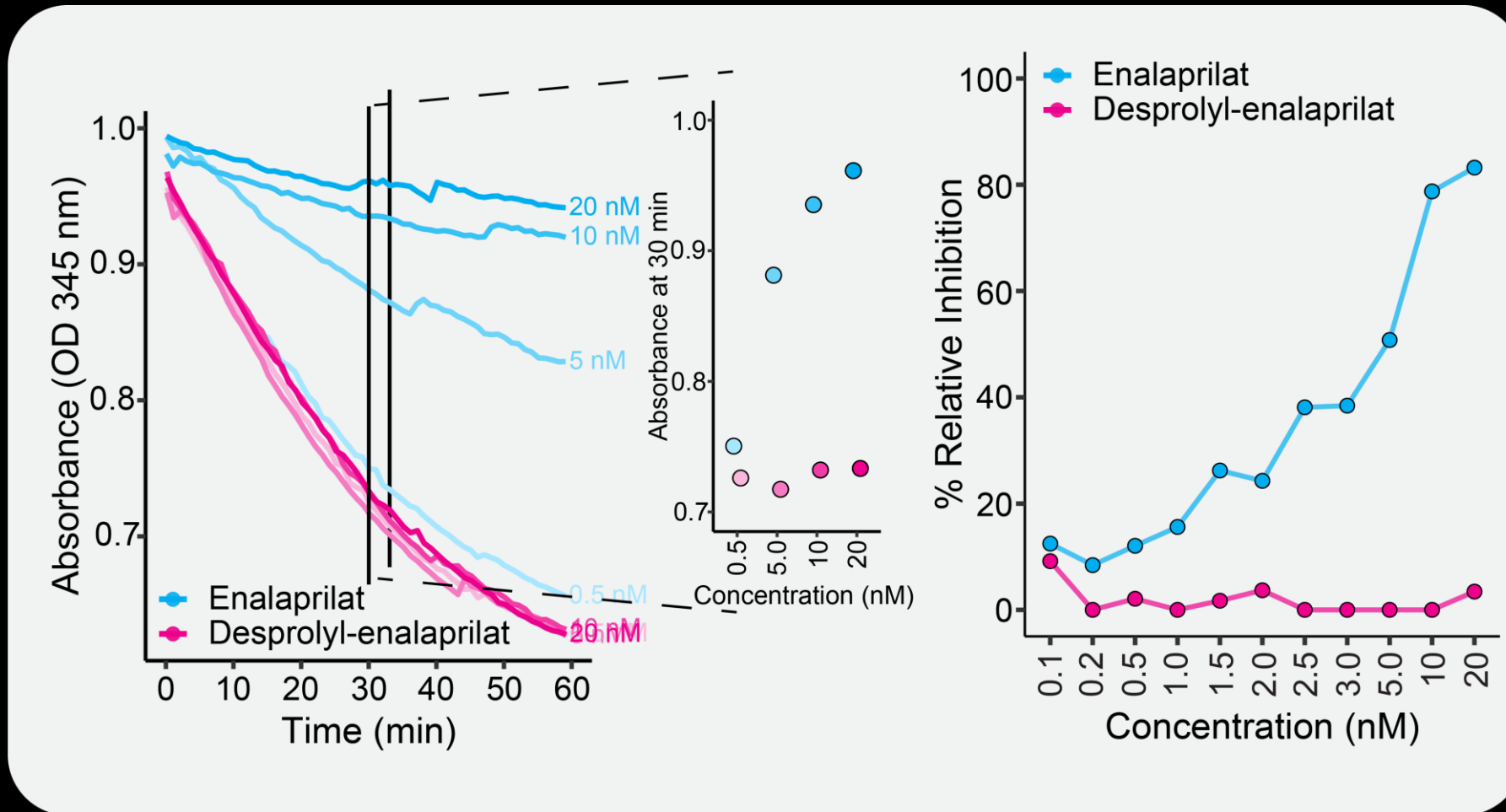
Protein
ACE1



Desprolyl-
enalaprilat

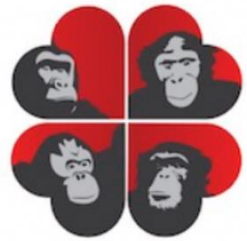
Desprolyl-enalaprilat has reduced H bonds potentially affecting ACE inhibition

Gut microbes inactivate the angiotensin-converting enzyme inhibitor drug enalaprilat



- 1- Does this microbiome-mediated inactivation translate to people?
- 2- Can we identify specific bacteria that do this?

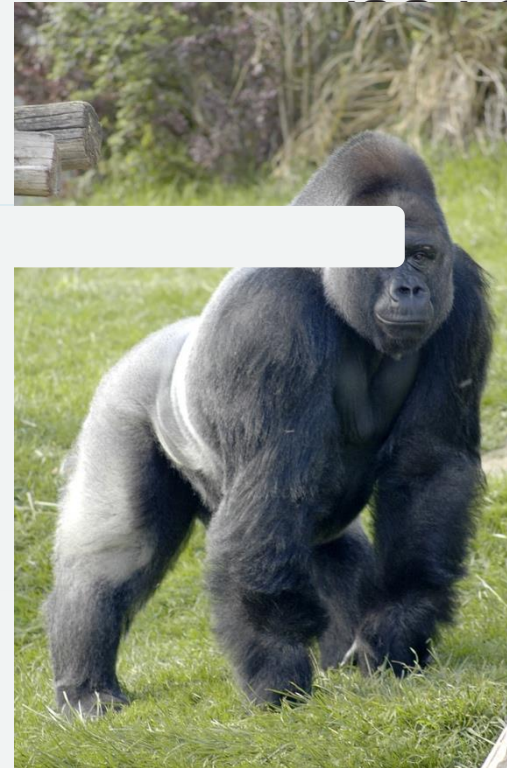
Contextualization of desprolyl-enalapril with microbiomeMASST



**THE GREAT APE
HEART PROJECT**
BASED AT DETROIT ZOO

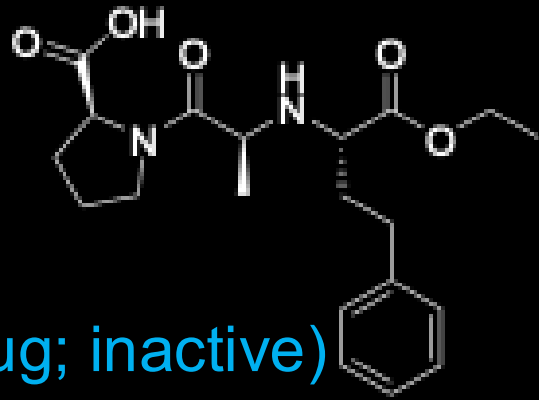
Therapeutic Considerations for the Treatment of Cardiovascular Disease in Great Apes

**CHRONIC HYPERTENSION WITH SUBSEQUENT CONGESTIVE
HEART FAILURE IN A WESTERN LOWLAND GORILLA (*GORILLA
GORILLA GORILLA*)**

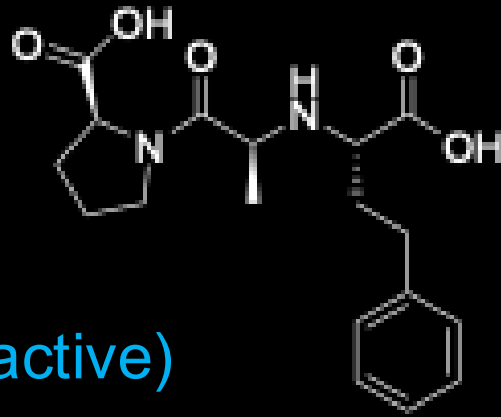


An angiotensin-converting enzyme inhibitor drug metabolite in Gorillas?

Desprolyl-enalapril only co-occur with enalapril

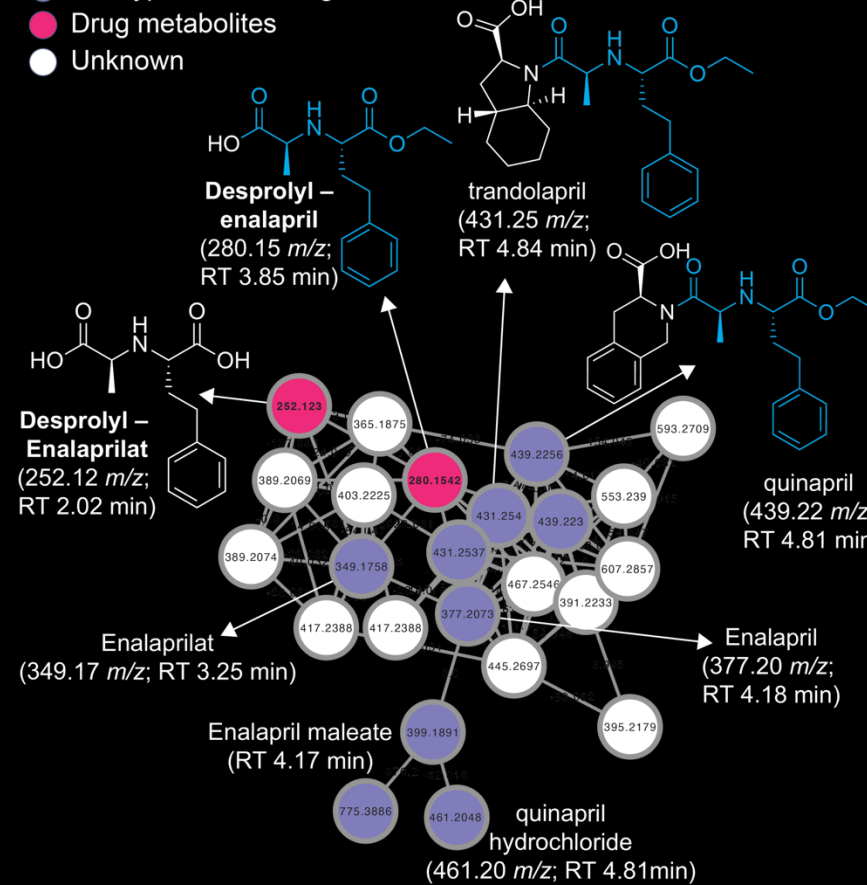


Esterases



Public dataset – human aging

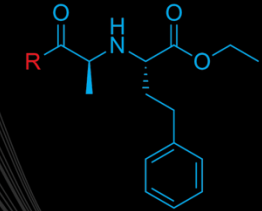
- Antihypertensive drugs
- Drug metabolites
- Unknown



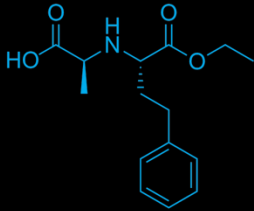
Public datasets (human, animal, bacteria)

MSV000079274
MSV000079598
MSV000080179
MSV000080186
MSV000080187
MSV000080673
MSV000081022
MSV000081492
MSV000081981
MSV000082711
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MSV000095728
MSV000095787
MSV000096049
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MSV000096589
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MSV000097935
MTBLS2746
MTBLS307
MTBLS417
NORMAN-6826b85a-74e4-55e5-b6e6-f656e50610aa
NORMAN-77e47d7d-7d29-5cde-97dd-698ce7395fd9
ST002044
ST002320

Parent drugs



Antihypertensive drug metabolite



enalapril

moexipril

quinapril

ramipril

trandolapril

Conclusions

- Discovery of a drug-activating function of microbial bile acid conjugation potentially influencing treatment outcomes.
- Leveraging reverse metabolomics to find new molecules, potential biomarkers, analyzing responses (responders vs non-responders), and even track environmental pollutants.
- Public data is a vast and yet underutilized resource.
- Harmonization and organization of these datasets enable scalable, microbiome-centric discovery and contextualization of microbial metabolites.

Charron-Lamoureux, V., et al. **A guide to reverse metabolomics-a framework for big data discovery strategy.** *Nat Protoc* (2025)

Charron-Lamoureux, V., [...], & Dorrestein, PC. **Pan-repository analysis reveals a drug-activating function of microbial bile acid conjugation.** *Under review at Nature*

Charron-Lamoureux, V., et al. **A searchable metadata network graph for microbiome metabolomics.** *Under review at Nature microbiology*

Patan, A*, Xing, S*, Charron-Lamoureux, V*, **Charting the Undiscovered Metabolome with Synthetic Multiplexing.** *Under review at Nature*

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Useful links – GNPS2 ecosystem

- https://gnps2.org/workflowinput?workflowname=everything_bagel_workflow
- https://apps.gnps2.org/sandbox/gnps2_networking_packager/
- <https://apps.gnps2.org/microbiomemasst>
- https://apps.gnps2.org/masst_search

 [Homepage](#) [Documentation](#) [About](#) [Workflows](#) [Tasks](#) [All Tasks](#) [File Browser](#)

GNPS2 Analysis Status Page

Clone Clone Hash Hide Task Delete Task Protect Task Public Task	Description	MSV000097665_BSH_EB_workflow	Update	Workflow Logs NF Report NF Timeline
	Task Tags		Update	
	Workflow	everything_bagel_workflow		
	Version	SERVER:2026.08.27;WORKFLOW:2026.09.14		
	Result Display	task View Latest Result Display		
	Status	DONE		
Task ID	5e6536ff7c704836873f91e6b77cd769			
User	Vince2212			

slurm (16)
[24/1eee70] Main:copyHTML | 1 of 1
[3a/045fe6] Main:listInputSpectra | 1 of 1
[df/d4192c] Main:buildLibraryCache | 1 of 1
[1d/2ea28f] Main:featureFinding | 1 of 1
[a1/54dfc1] Main:buildWorkflowSummary (1) | 1 of 1
[cd/163b32] Main:fillMGFScans (1) | 1 of 1
[6d/657bf2] Main:librarySearchRust (1) | 1 of 1
[9f/bde90d] MaieatureLibrarySearchResults | 1 of 1
[d6/c32d3e] Main:loadLibrarySearchRust (1) | 1 of 1

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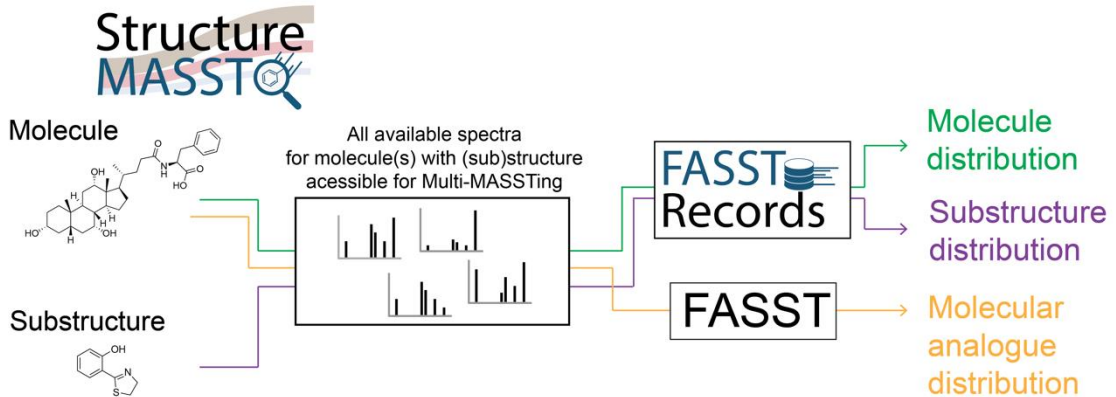
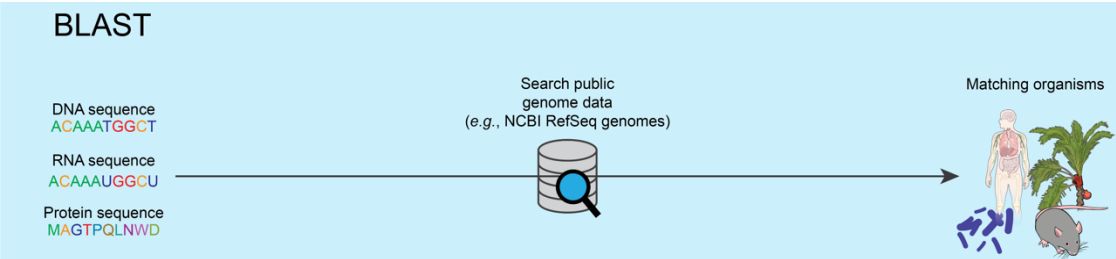
Structure-centric searching enables global mapping of the public metabolome

[Yasin El Abiead](#) , [Jeong In Seo](#), [Vincent Charron-Lamoureux](#), [Michael Strobel](#), [Wilhan Donizete Gonçalves Nunes](#), [Haogi Nina Zhao](#), [Kine Eide Kvitne](#), [Simone Zuffa](#), [Helena Mannocho-Russo](#), [Harsha Gouda](#), [Cristina Bez](#), [Abubaker Patan](#), [Shipei Xing](#), [Jasmine Zemlin](#), [Ipsita Mohany](#), [Julius Agongo](#), [Andres Mauricio Caraballo Rodriguez](#), [Lindsey A. Burnett](#), [Victoria Deleray](#), [Abzer K. Pakkir Shah](#), [Jarmo-Charles Kalinski](#), [Daniel Petras](#), [Nikiforos Alygizakis](#), [Jeremy Carver](#), ... [Pieter C. Dorrestein](#)  [+ Show authors](#)

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<https://structure-masst.gnps2.org/>



structureMASST

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geoMASST (under construction)

lifeMASST (under construction)

Contributors

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Mingxun Wang (UCR)

Session: 41e4da21bc66

Type a chemical name to search PubChem

e.g., diazepam, caffeine, etc.

or

SMILES/SMARTS



Enter SMILES or SMARTS

or

 Add batch file 

Find available MS/MS spectra

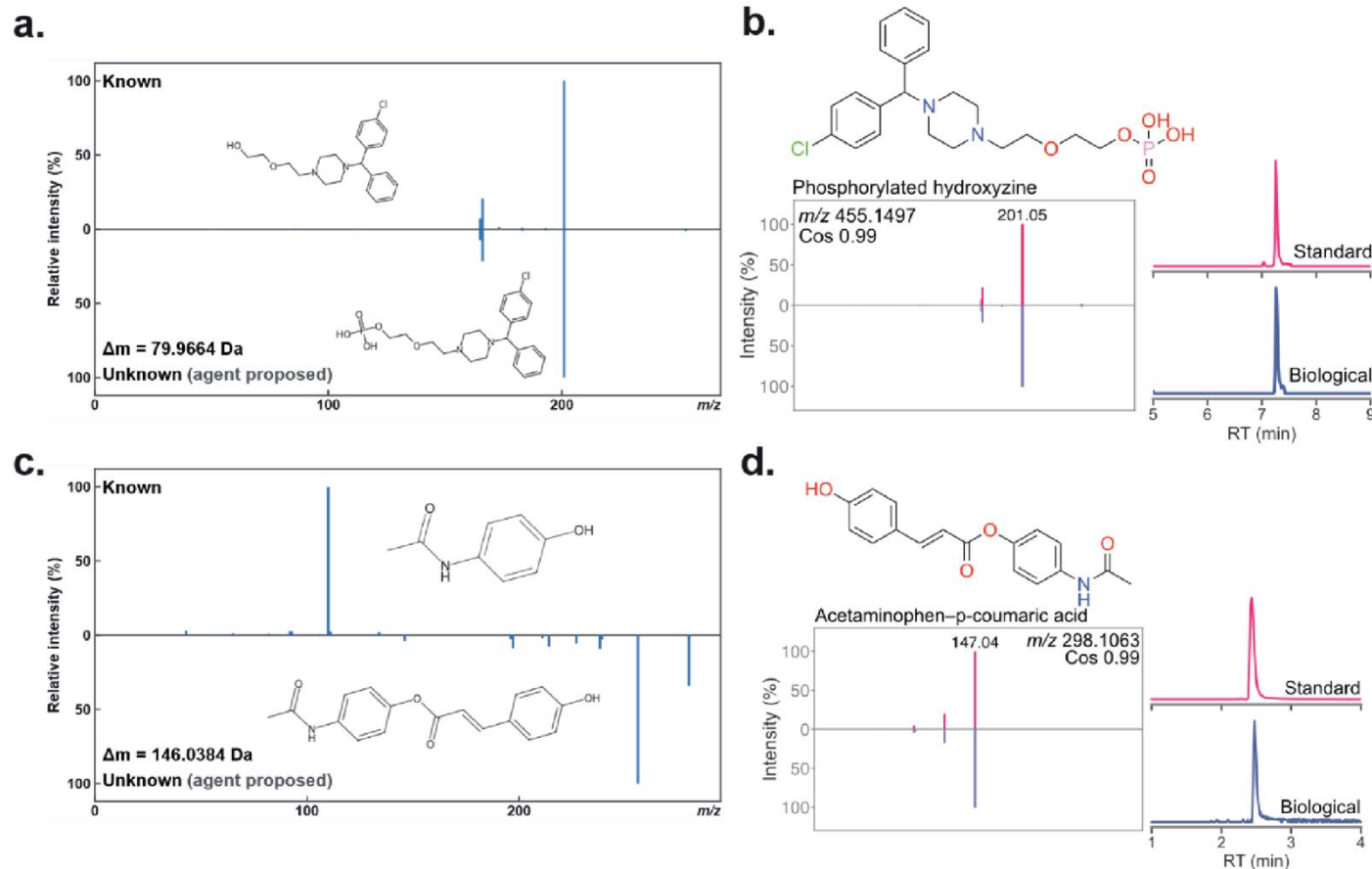
☒ Exact structure match ☐ Substructure match ☐ Tanimoto similarity

 Get Available Spectra

Agentic AI for Structural Elucidation and Discovery of Drug Metabolites from Mass Spectrometry Data

 Xianghu Wang,  Abubaker Patan, Haoqi Nina Zhao,  Vincent Charron-Lamoureux, Yourae Shin, Daniel Petras,  Yuhui Hong,  Benjamin P. Bowen,  Trent R. Northen,  Pieter C. Dorrestein,  Mingxun Wang

doi: <https://doi.org/10.64898/2026.06.23.734138>



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doi: <https://doi.org/10.64898/2026.08.17.744735>

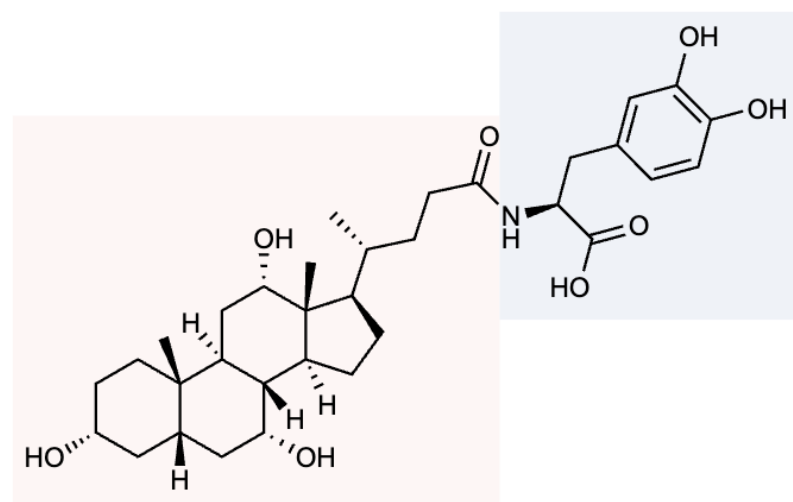
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Navigating the conjugated metabolome

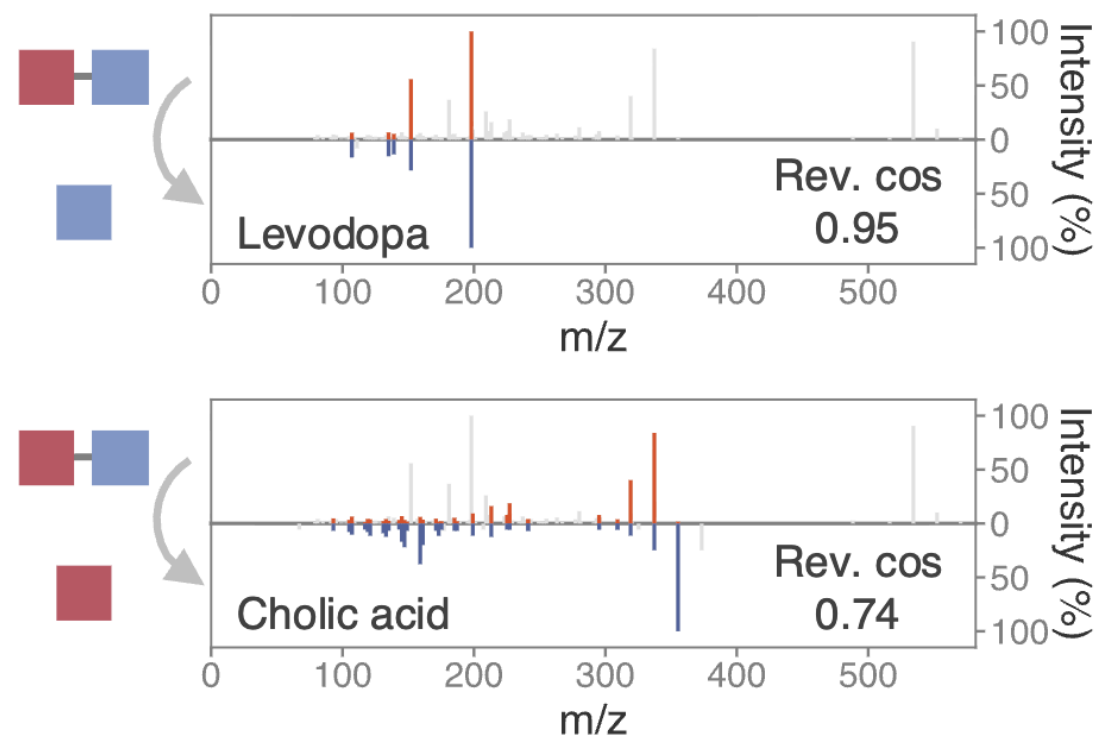
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doi: <https://doi.org/10.64898/2026.02.06.704496>

b



Choly-levodopa



Navigating the conjugated metabolome

Shipei Xing, Abubaker Patan, Julius Agongo, Harsha Gouda, Vincent Charron-Lamoureux, Yasin El Abiead, Zhewen Hu, Haoqi Nina Zhao, Ipsita Mohanty, Jasmine Zemlin, Wilhan Donizete Gonçalves Nunes, Lindsey A. Burnett, Mingxun Wang, Dionicio Siegel, Pieter C. Dorrestein

doi: <https://doi.org/10.64898/2026.02.06.704496>

